Waterbouwkunde
CTB2410

Week 7

Hydro-energy

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8 June, 2026

TU Delft



Hydro-energy

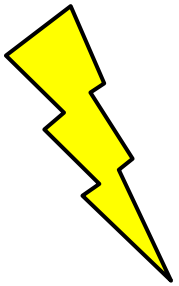
Today's topics

- Hydropower's role in the energy mix
- Electricity cost and price
- Energy Transformation
- Hydropower schemes
  - High head
  - Reservoir
  - Run of river
  - Pumped storage
  - small hydropower
  - Tidal barrage
- Turbines
- Reservoirs and dams

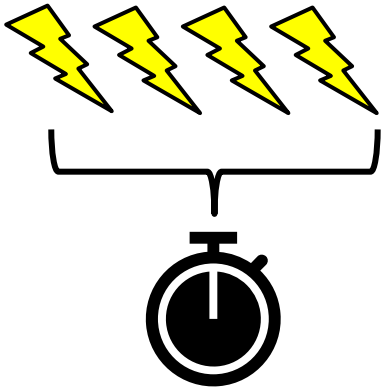
Energy in the world

Power vs. Energy

Power: The instantaneous output (Voltage X Current) from a generator, in kW (in hydropower engineering). In hydropower, is understood as the maximum power that can be generated.



Energy: The amount of generated (or consumed) power over time in kW-hrs (in hydropower engineering).

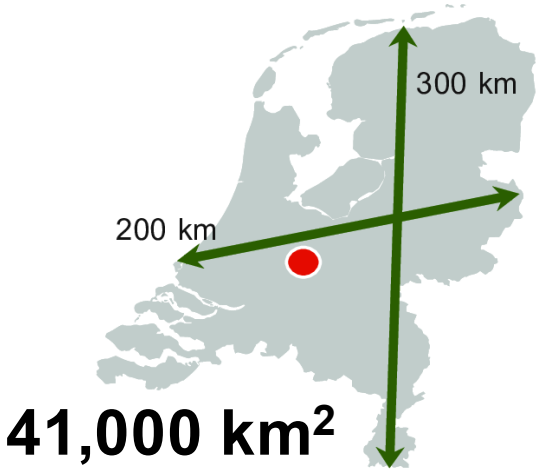


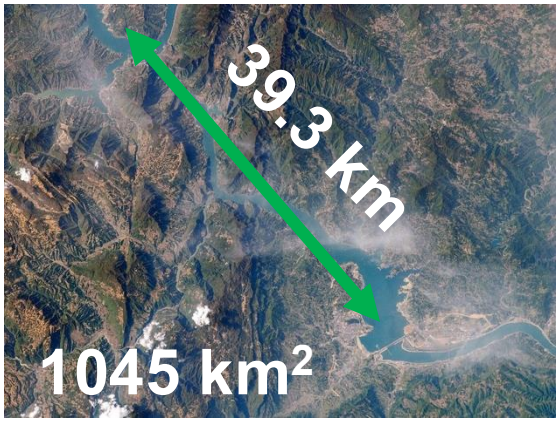
Netherlands with the 3 Gorges Dam

Dutch total E-consumption 2023 is ~ 109 000 GWh/yr

Simple, but risky !?

Three Gorges Dam has an installed capacity of P=22,500 MW
If production ~55%*24h/d*365 d/yr , then E ~ 100 000 GWh/yr



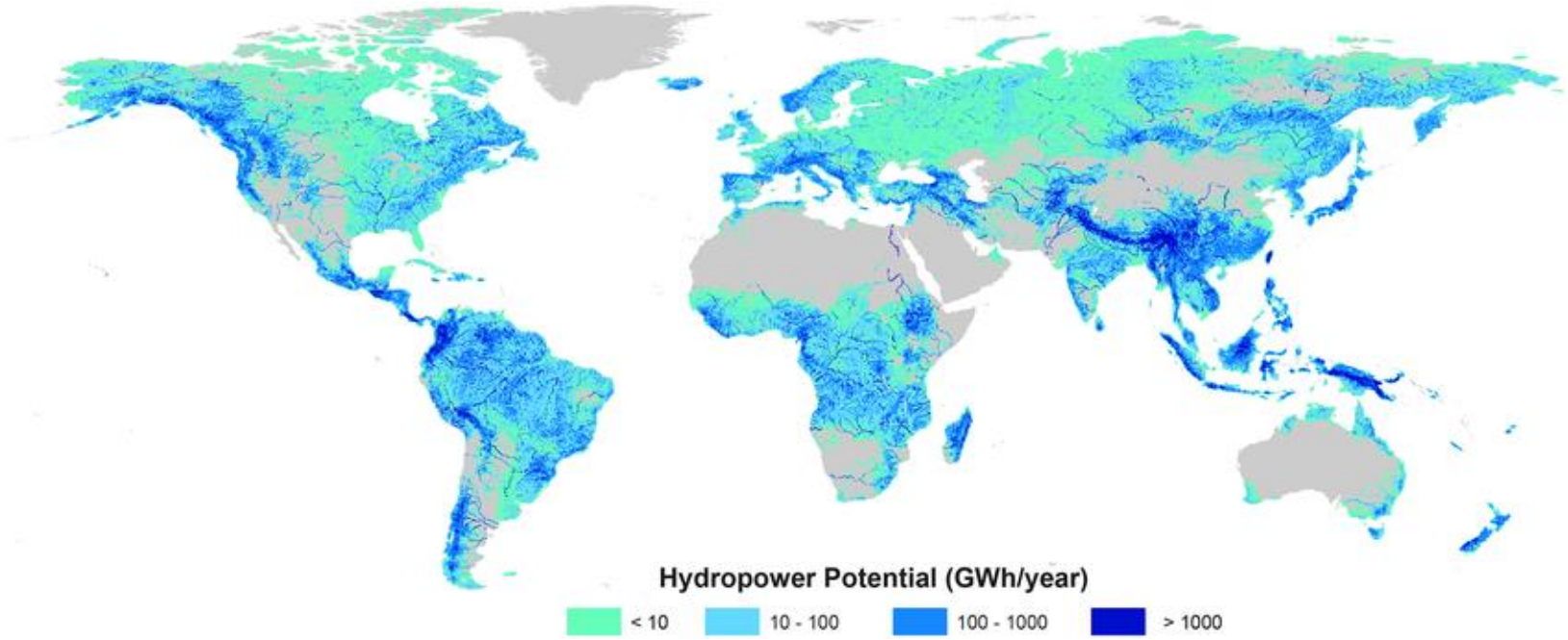


2.55% !

Image from Phd Thesis: ISBN 978-94-6233-529-5 TRAUMA SYSTEMS AN ERA OF DEVELOPMENT

http://floodlist.com/asia/china-over-1-million-affected-by-floods-in-hubei

Global Hydropower Potential



Hydropower potential is the theoretical maximum amount of electricity that can be generated from a specific area's water resources, primarily determined by the water flow (volume of water) and the working head (vertical distance the water falls).

How much do you pay for kW-h ?

1. I pay ~ 0.05 €/kW-h
2. I pay ~ 0.10 €/kW-h **Cost of Production**
3. I pay ~ 0.15 €/kW-h
4. I pay ~ 0.20 €/kW-h
5. I pay ~ 0.30 €/kW-h **Price for Consumption**

Question: How much does it cost to charge your mobile per year ?

Answer: Average iPhone or Android battery holds one charge of about 15 watt-h.

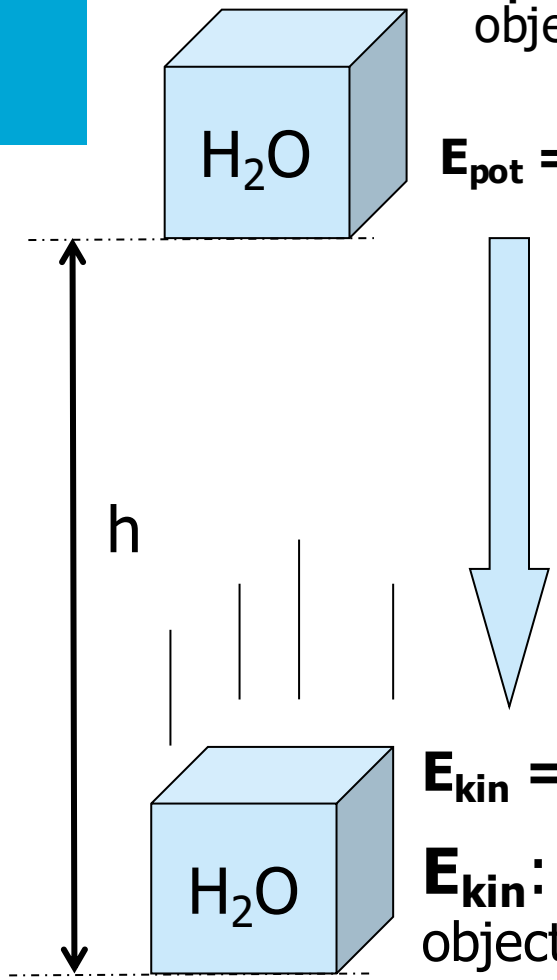
Charge (365 times) : 5.5 kW-h/year

Total cost: 1.65 €/year



Energy Generation (Physics 101)

Energy transformation



E_{pot}: Energy "stored" in an object due to its position.

E_{pot} =m*g*h [$kg.m^2/s^2$]

= [N.m]

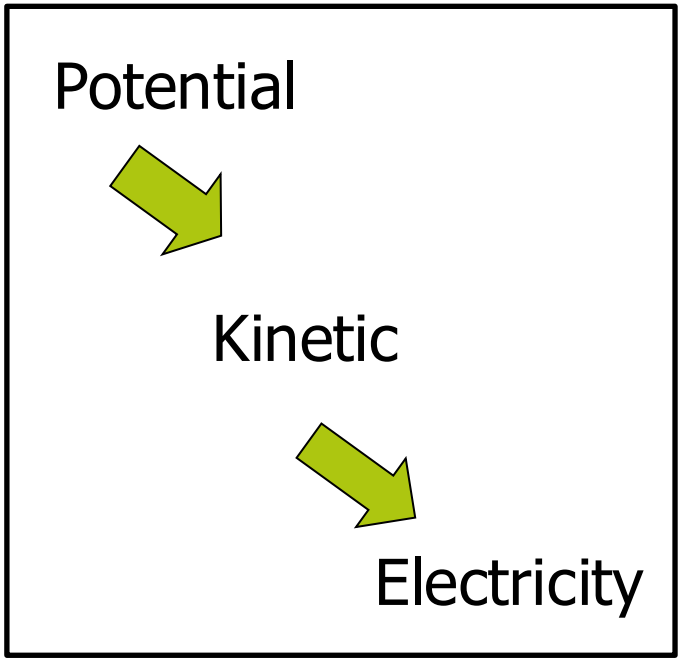
= [J]

= [W.s]

E_{kin} =1/2*m*v^2

E_{kin}: Energy a moving object has due to motion.

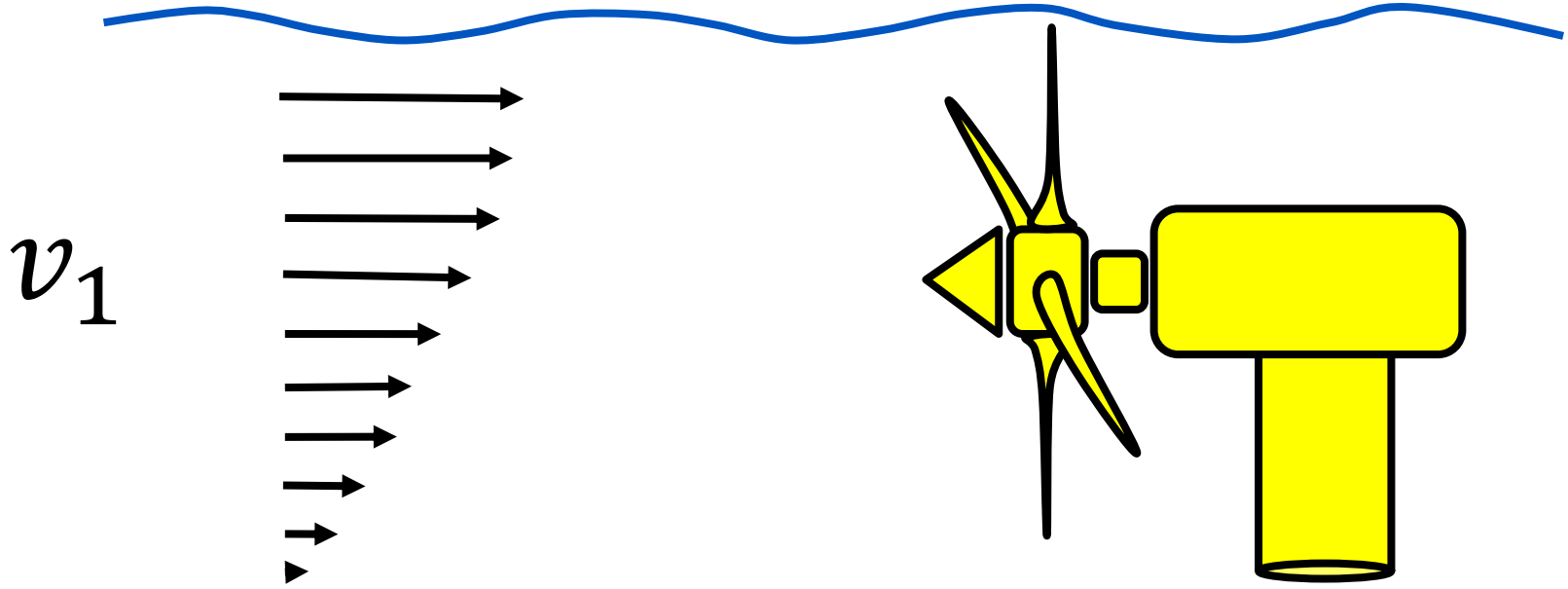
$$E_{pot} + E_{kin} = Const.$$



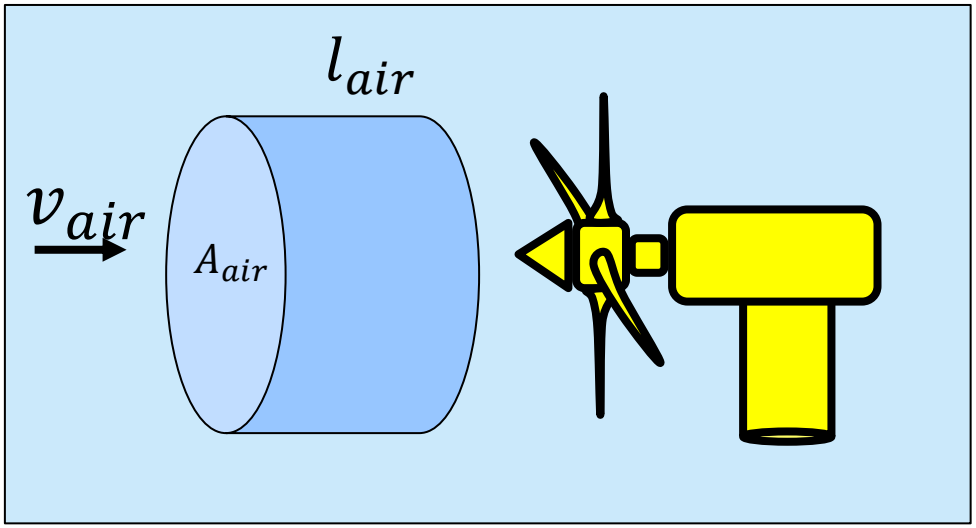
Can you derive the power equation of a kinetic stream waterpower system ?

Tip: Power is energy generated per unit of time. Bernoulli's principle expresses the energy per UNIT OF VOLUME.

$$P[Watts] = ?$$



Answer: Kinetic stream power equation



$$E_{kin} = \frac{1}{2} m_{air} \, {v_{air}}^2 [W.s]$$

$$\rho_{air} = \frac{m_{air}}{A_{air} * l_{air}} \left[\frac{kg}{m^3}\right]$$

$$E_{kin} = \frac{1}{2} (\rho_{air} * l_{air} * A_{air}) \, {v_{air}}^2 [W.s]$$

E_{kin}: Energy a moving object has due to motion.

$$\frac{E_{kin}}{t} = \frac{1}{2} \frac{(\rho_{air} * l_{air} * A_{air}) \, {v_{air}}^2 [W.s]}{t[s]} [W.s]$$

$$P_{kin} = \frac{1}{2} (\rho_{air} * A_{air}) \, {v_{air}}^3 [W]$$

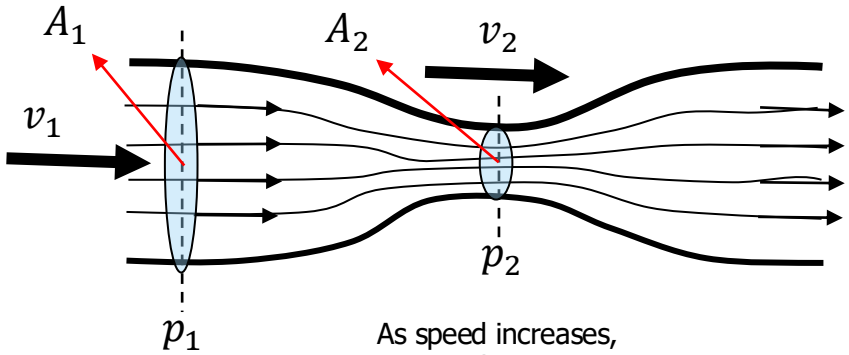
Bernoulli's Principle (Euler's Equation)

$$H_T\left[\frac{N}{m^2} * \frac{1}{m^3}\right] = p_1 + \rho_w g h_1 + \frac{\rho_w {v_1}^2}{2} = p_2 + \rho_w g h_2 + \frac{\rho_w {v_2}^2}{2}$$

Pressure energy ($\mathbf{E_p}$)	Potential energy per unit of volume ($\mathbf{E_{pot/[vol]}}$)	Kinetic energy per unit of volume ($\mathbf{E_{kin/[vol]}}$)

Continuity equation

$$v_1 A_1 = v_2 A_2$$



$A_2 < A_1$

$v_2 > v_1$

$p_2 < p_1$!

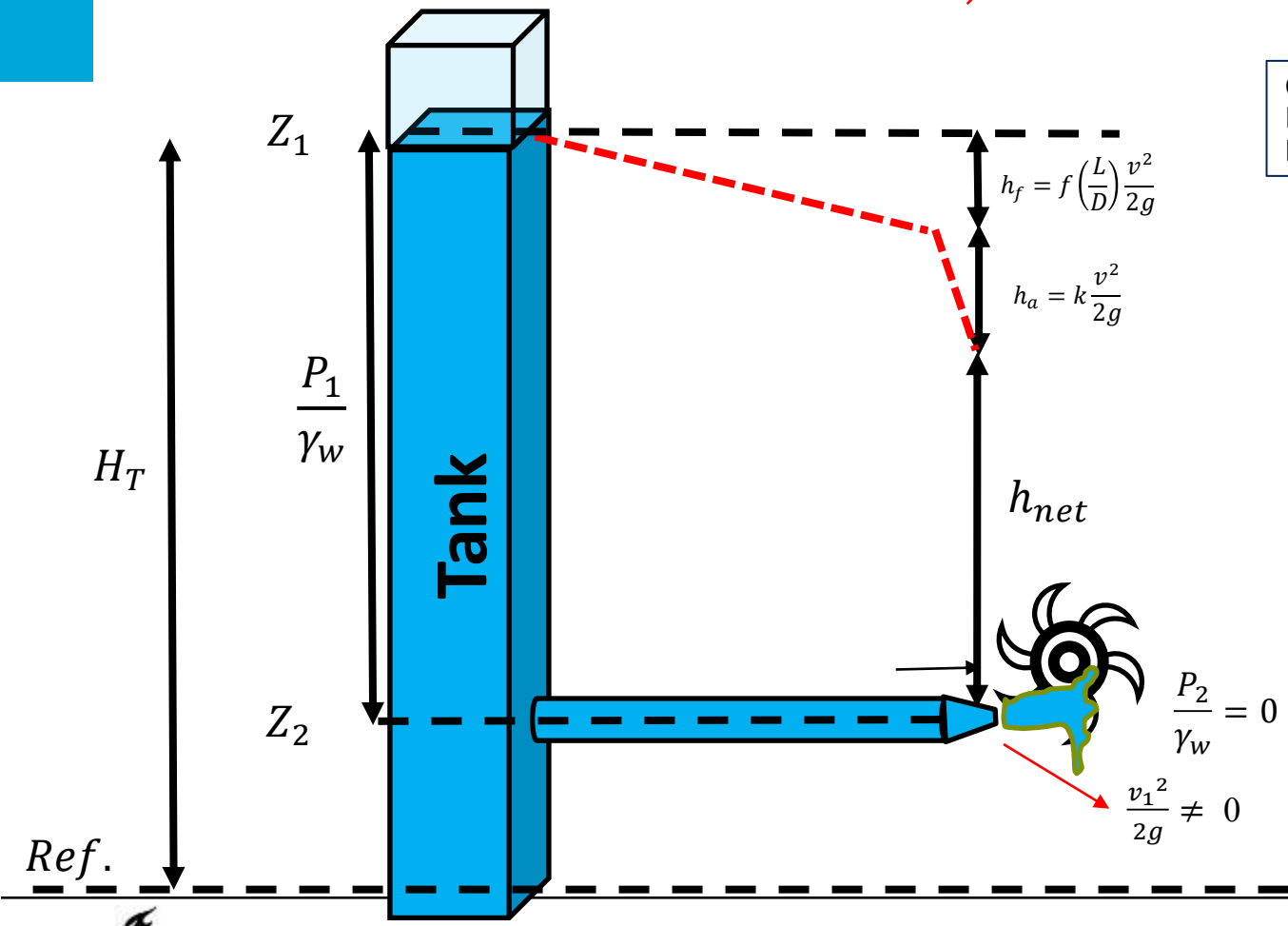
" The increase in fluid speed will occur simultaneously with a decrease in static pressure and/or a decrease in its potential energy."

Bernoulli's Equation (with losses)

$$H_T[m] = Z_1 + \frac{P_1}{\gamma_w} + \frac{{v_1}^2}{2g} = Z_2 + \frac{P_2}{\gamma_w} + \frac{{v_2}^2}{2g} + h_f + h_a$$

Localized Friction loses

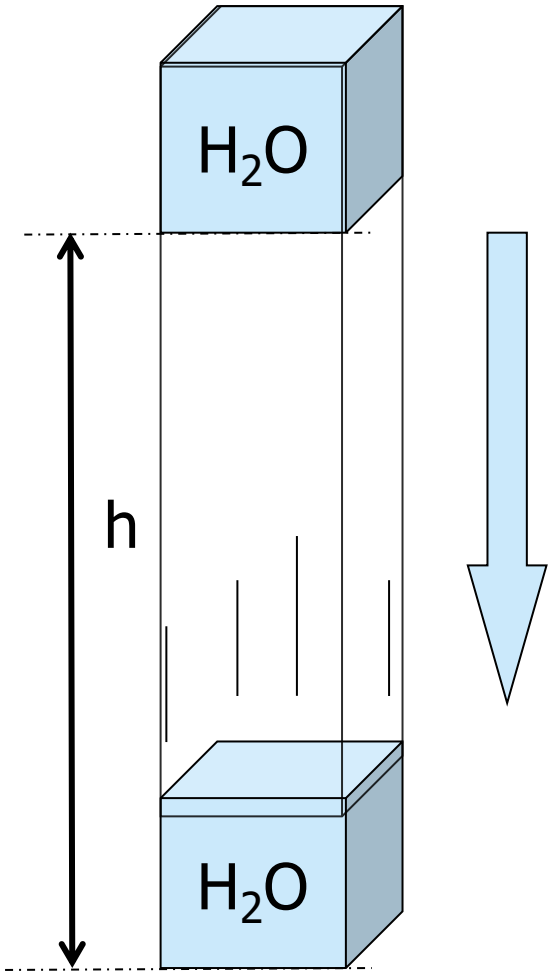
Conduit Friction loses



h_{net}: Available energy that can be used for generation in the turbine after losses.

Loses are due to friction and they are transformed into thermal energy (heat !)

Turbine power from work (Energy transfer)



$$W[N \cdot m] = F \times d$$

$$W[J] = \left(m[kg] * g\left[\frac{m}{s^2}\right]\right) h[\mathrm{m}]$$

$$P[Watts] = \frac{W[J]}{t[s]} = \frac{(m * g)\, h}{t}$$

$$P[Watts] = \frac{\left(\rho_w \left[\frac{kg}{m^3}\right]\right)(m * g)\, h}{\left(\rho_w \left[\frac{kg}{m^3}\right]\right) t}$$

$$P[Watts] = Q\ h\ g\rho_w$$

But … Turbines have losses too (η) !



https://www.asme.org/topics-resources/content/lester-allan-pelton



https://commons.wikimedia.org/wiki/Francis-Turbine



https://www.zeco.it/zeco-turbines/kaplan-turbine/

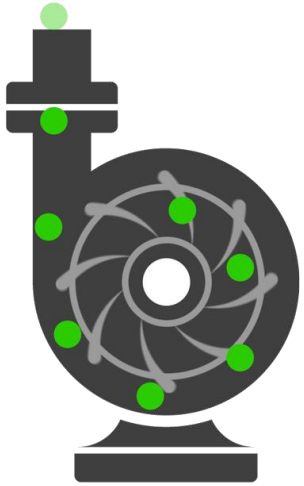
$$P[Watts] = \eta\ Q\ h_{net}\ g\rho_w\ !!!$$

Hydropower Equation (Mechanical)

And … Pumps too (η) !

$$P[Watts] = Q\ h_{net}\ g\rho_w / \eta\ !!!$$

Pump Equation (Mechanical)



Hydropower schemes and main structures

Hydropower scheme:

Is the combination of civil, mechanic and electric elements and their location in space required to generate the available hydropower potential given a specific location.



Source: https://www.renewableenergyworld.com/baseload/hydropower/thailand-planning-massive-floating-solar-power-plants-on-hydropower-dam-reservoirs/#gref

Inland hydropower plant classification based on installed capacity

Large (> 100 MW)



Medium (10 MW - 100 MW)



Small (1 MW - 10 MW)



Mini (100 kW – 1 MW)



Micro (1 kW – 100 kW)

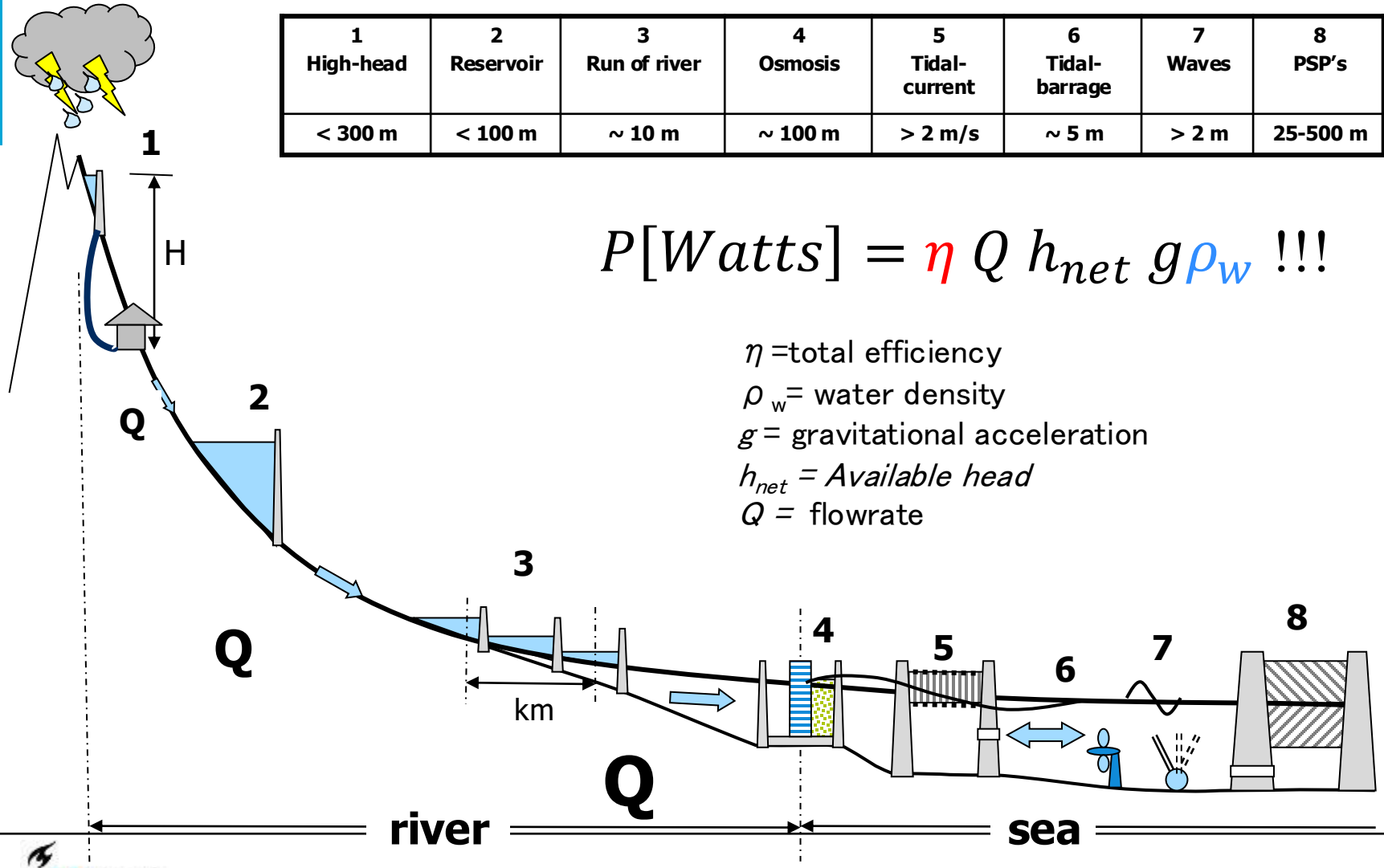


Pico (< 1 kW)



Hydropower plant classification based on head

1 High-head	2 Reservoir	3 Run of river	4 Osmosis	5 Tidal-current	6 Tidal-barrage	7 Waves	8 PSP's
< 300 m	< 100 m	~ 10 m	~ 100 m	> 2 m/s	~ 5 m	> 2 m	25-500 m

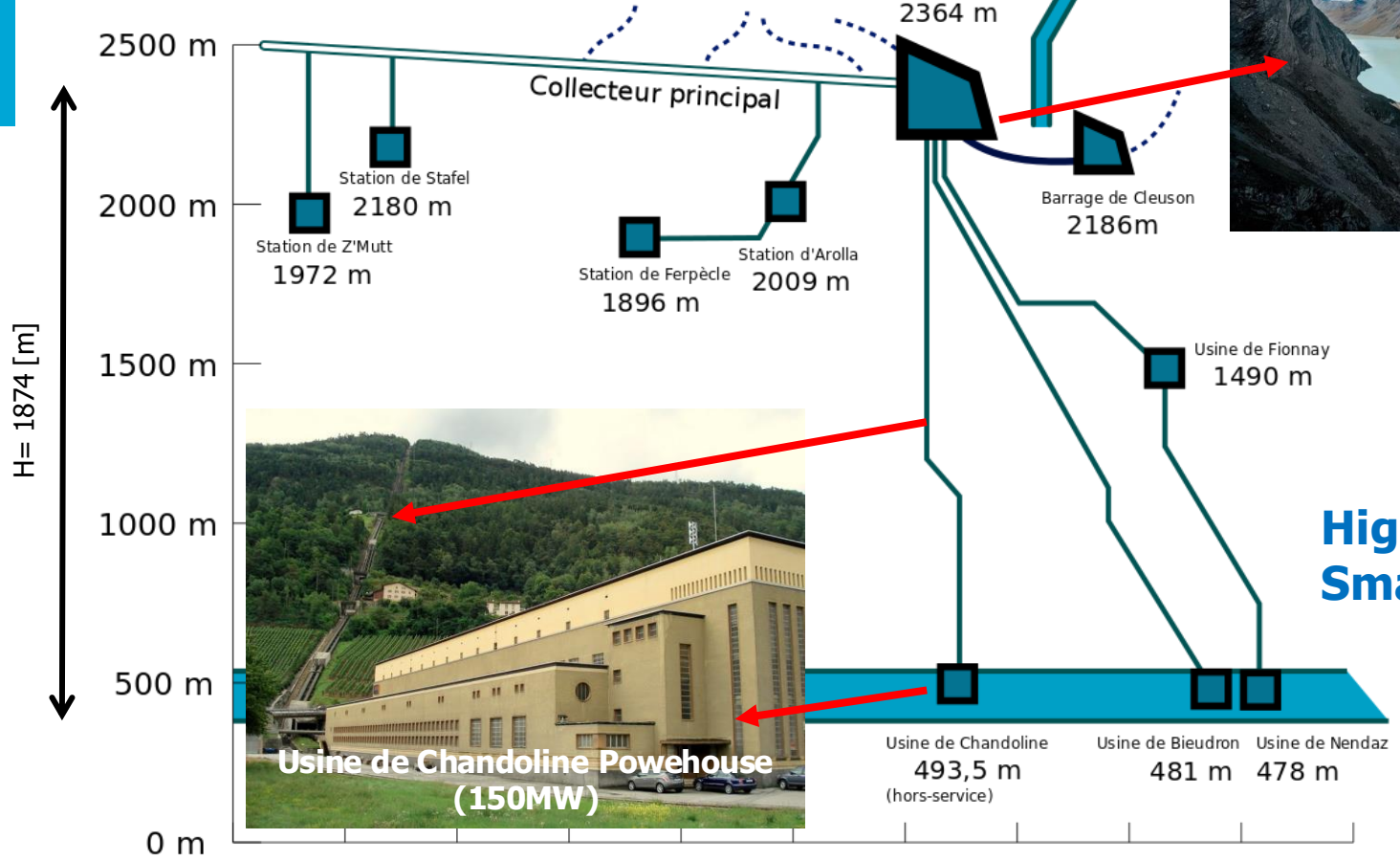


$$P[Watts] = \eta\, Q\, h_{net}\, g \rho_w \;!!!$$

η =total efficiency
ρ_w= water density
g = gravitational acceleration
h_{net} = *Available head*
Q = flowrate

1. High head power plants



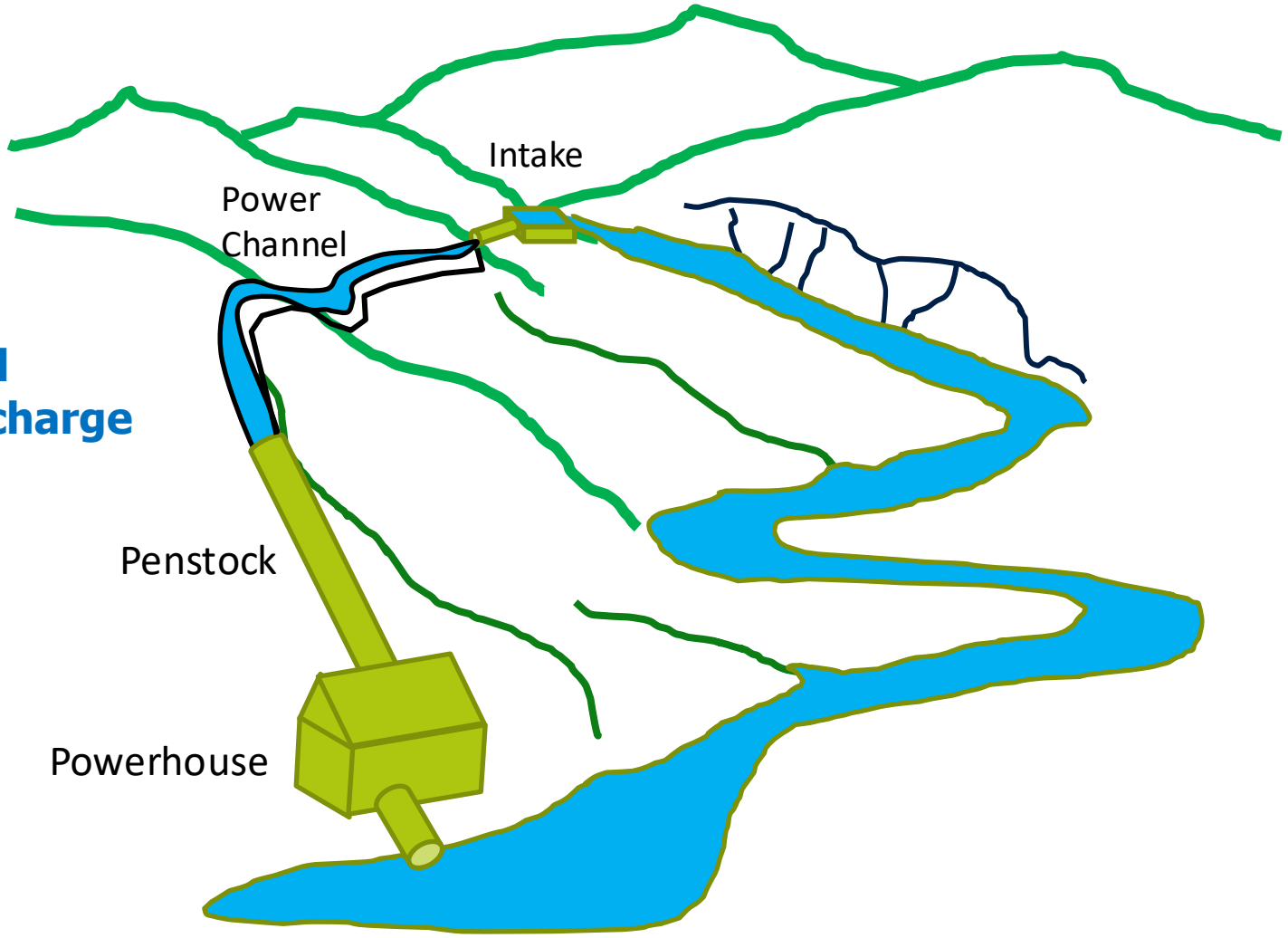


High Head
Small Discharge

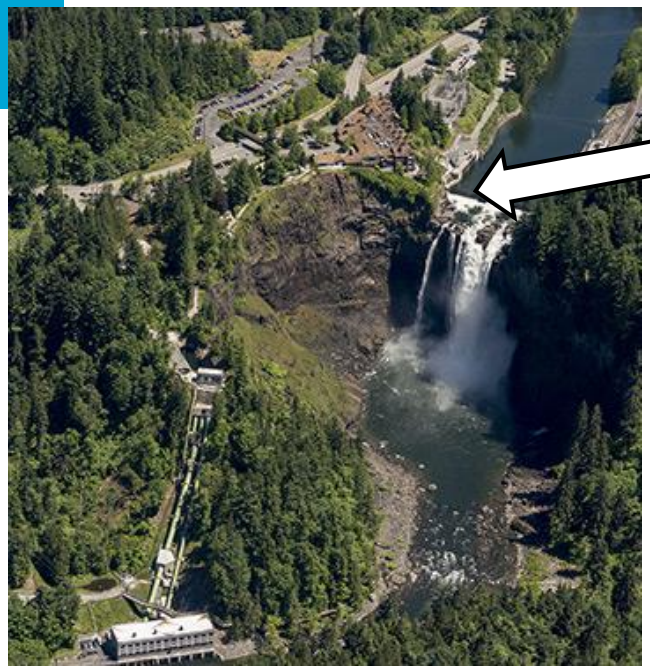
http://www.grande-dixence.ch/

Scheme: Open power channel (run-of-river)

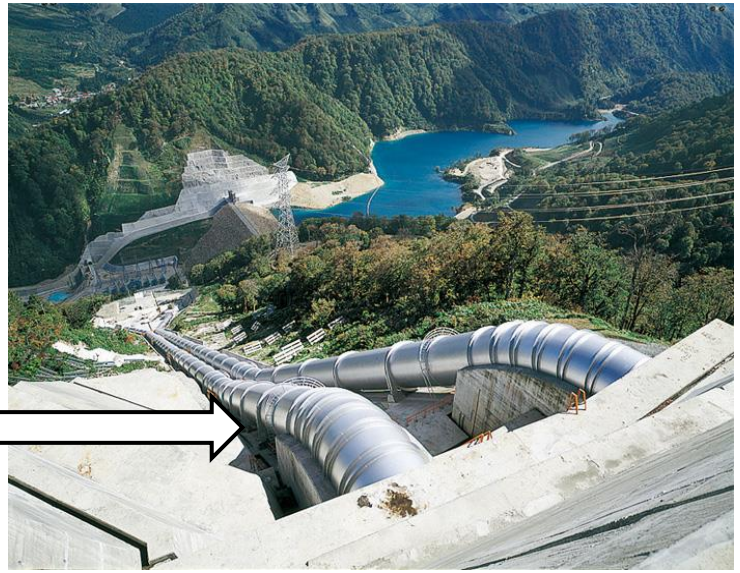
High Head
Small Discharge

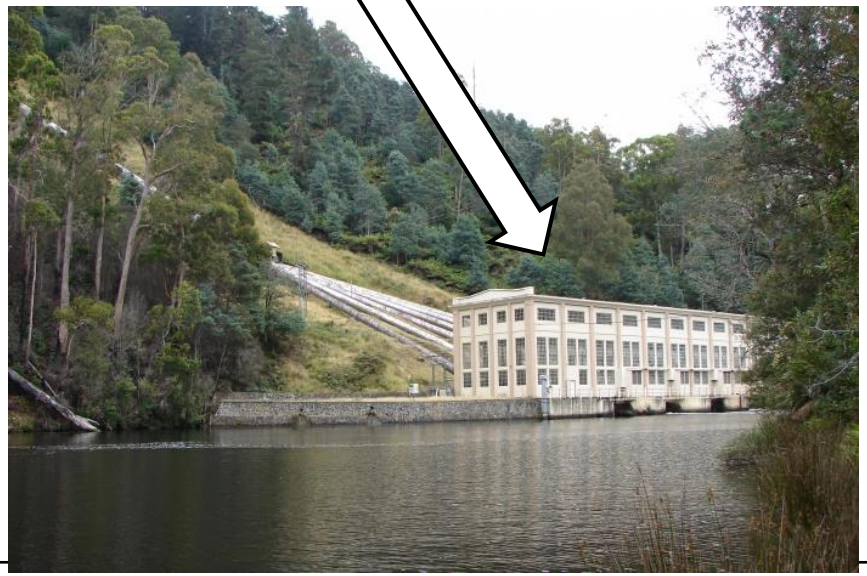


High Head Power Plant



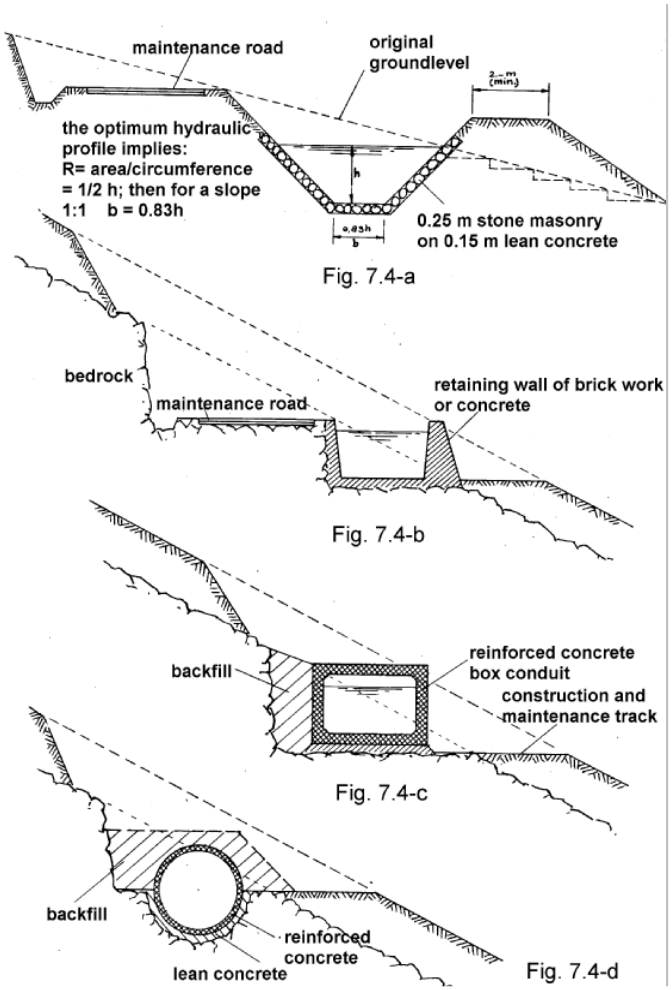
Intake,
Penstock
& Powerhouse





Power channel (headrace)





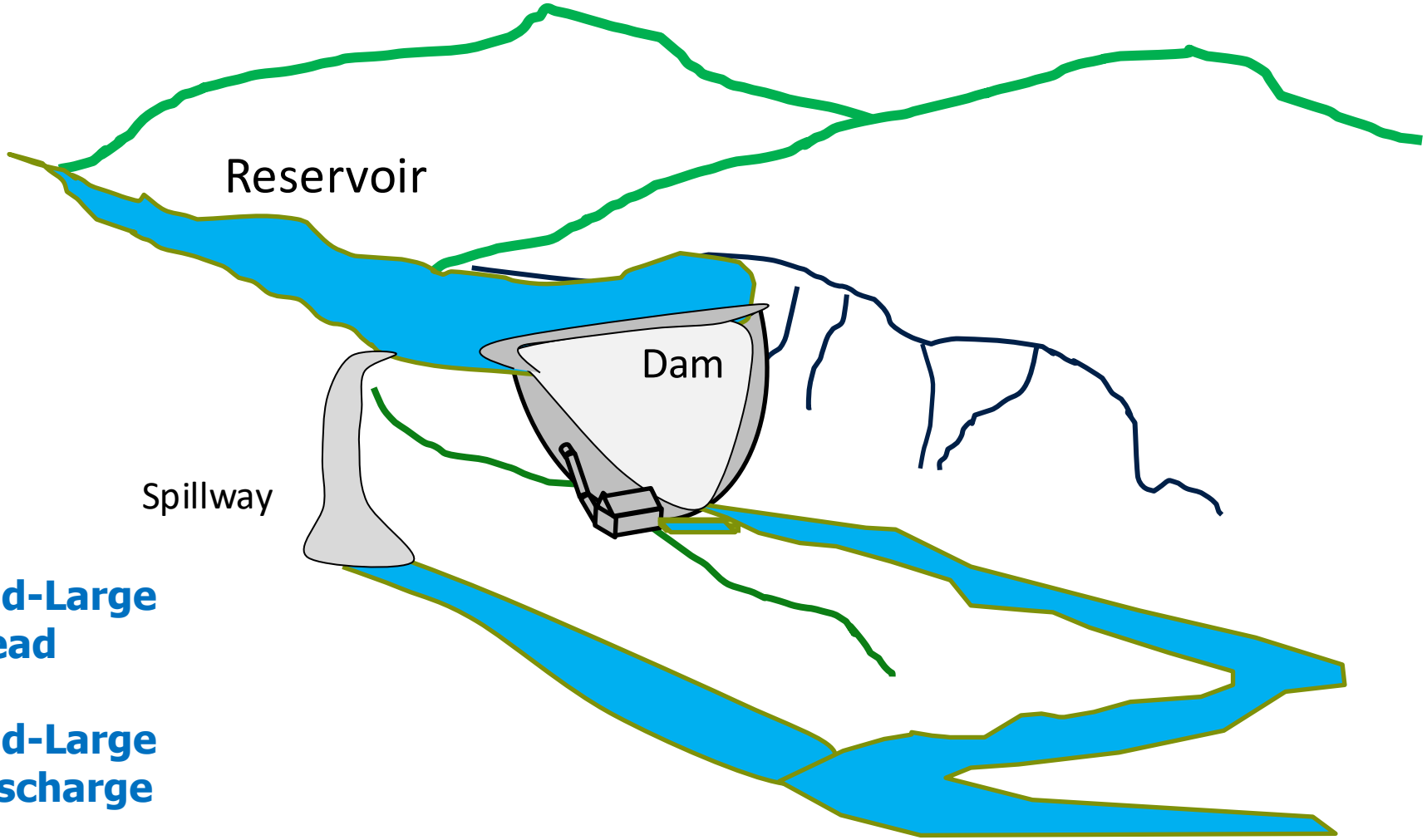
2. Reservoir power plants

Mid-Large Head

Mid-Large Discharge



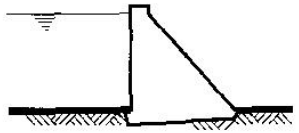
Scheme: Storage and damming



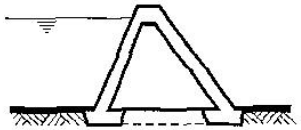
Mid-Large Head

Mid-Large Discharge

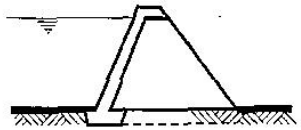
Dams



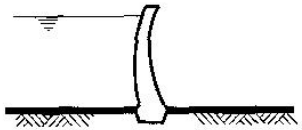
Concrete/Masonry Gravity



Hollow Gravity

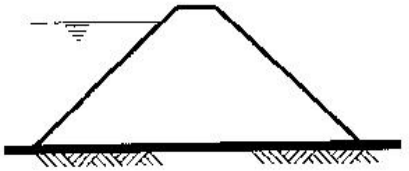


Concrete Buttress

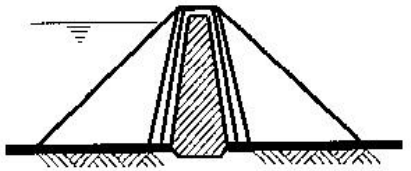


Concrete Arch

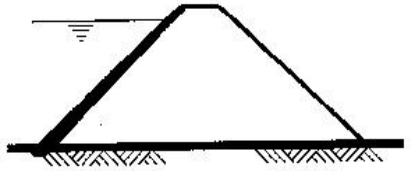
Homogenous Embankment

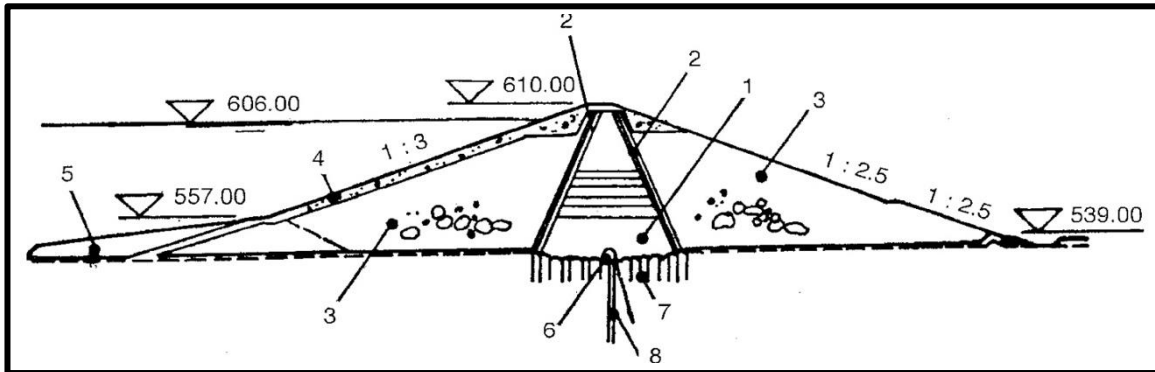


Zoned Embankment



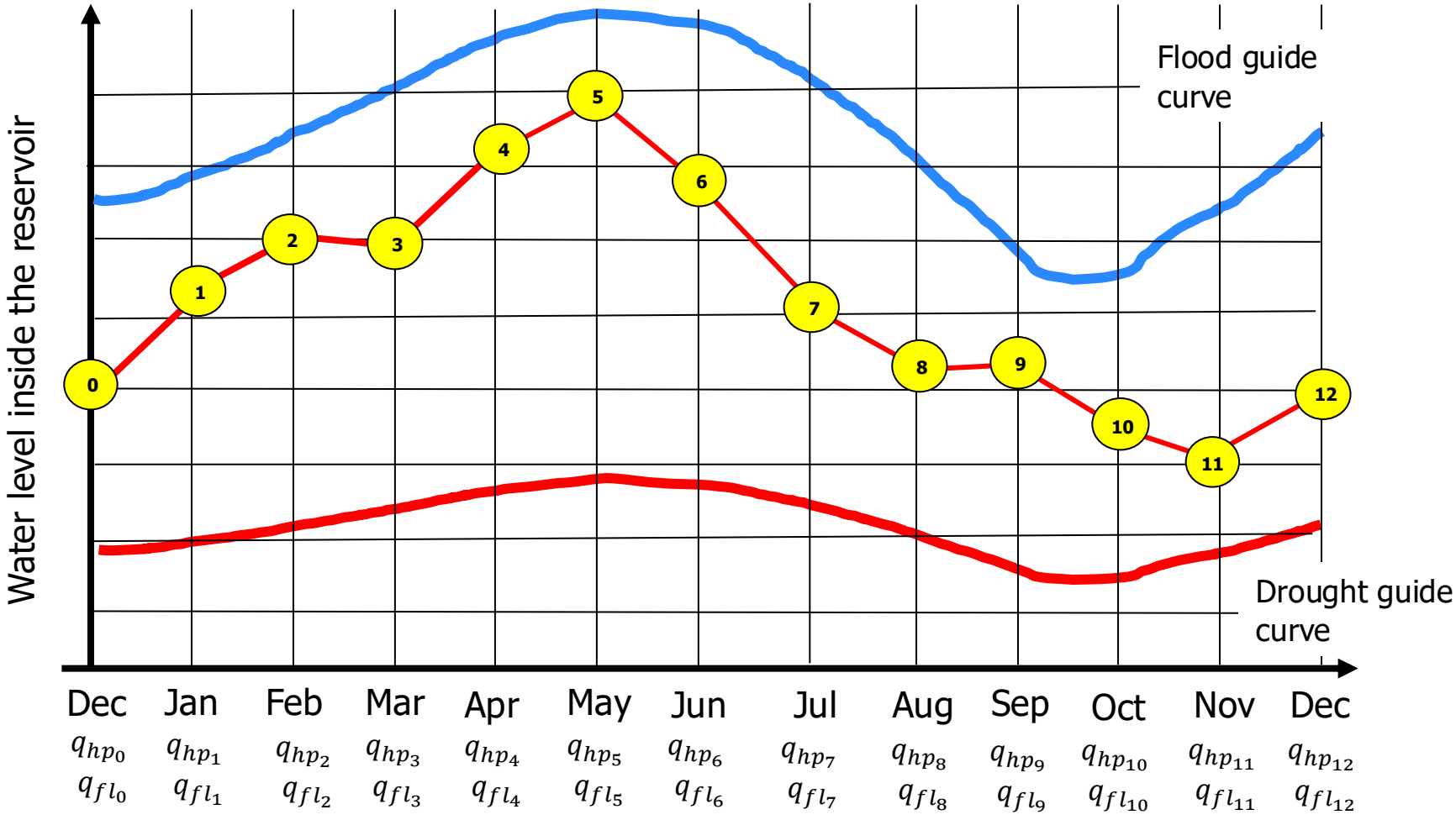
Embankment with Waterproof Facing





1 – clay core;
2 – filter;
3 – sand and gravel shell;
4 – rockfill protection;
5 – cofferdam;
6 – grouting and drainage gallery;
7 – consolidation grouting;
8 – grout curtain.

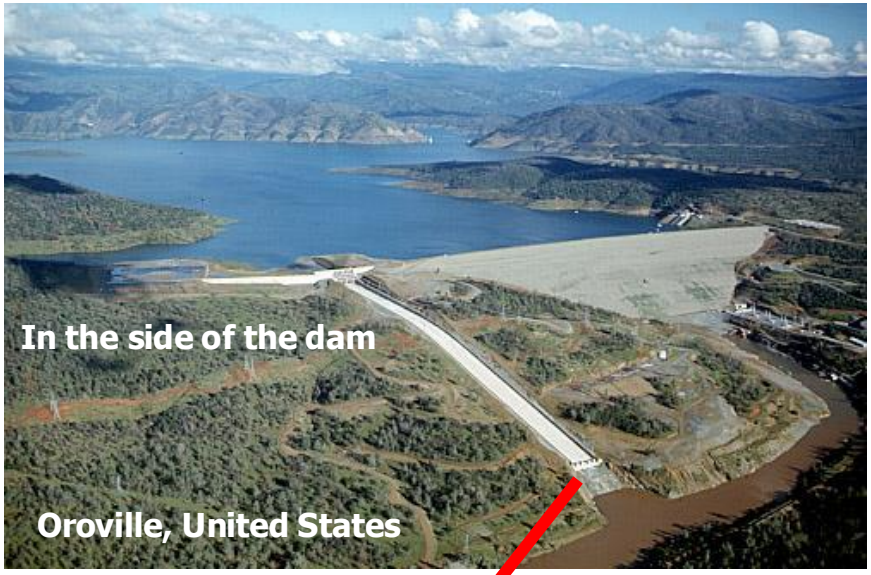
Reservoir operation optimization



Spillway (diversion)

Spillway: extreme value analysis?





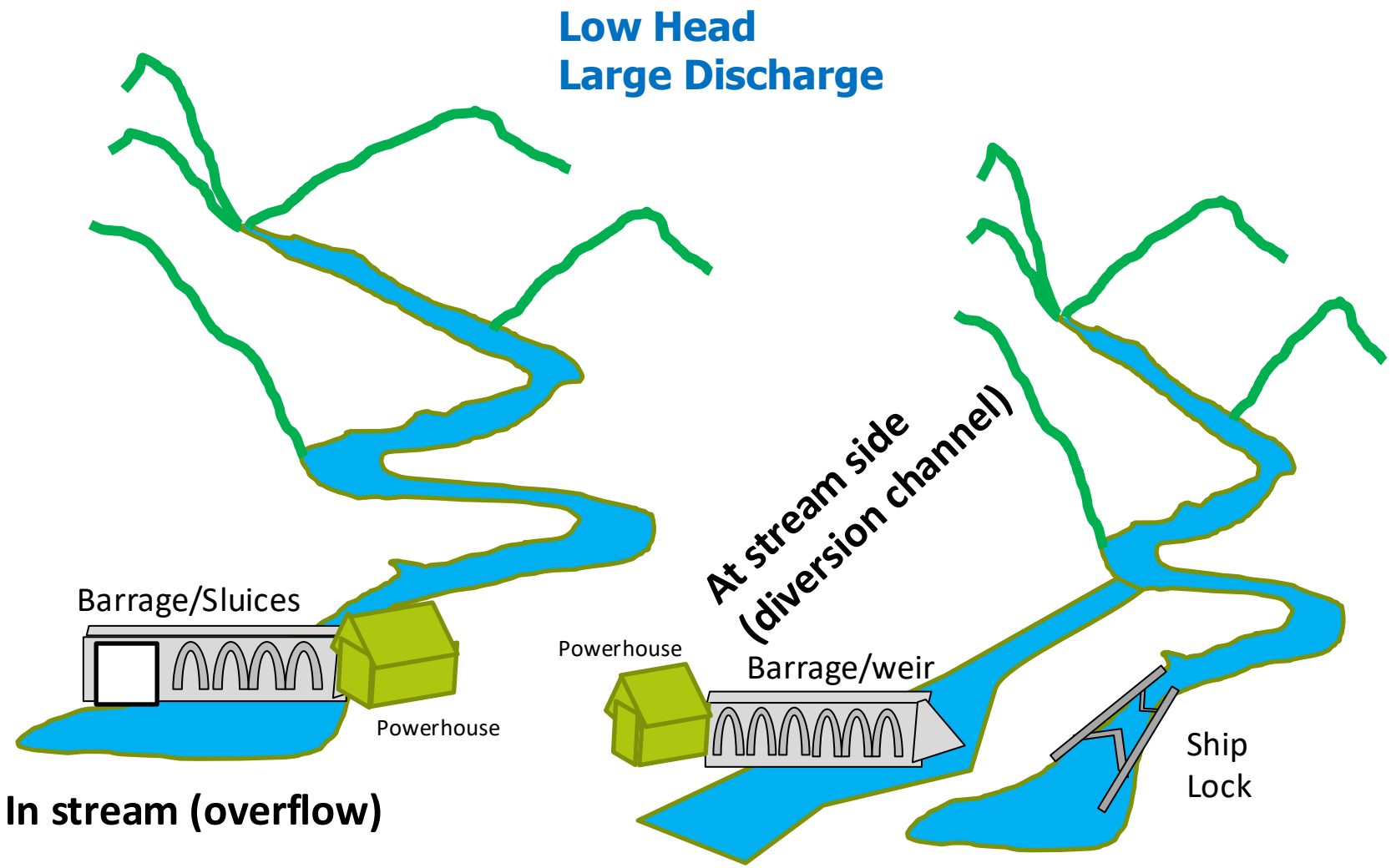




3. Run-of-river plants (low head)

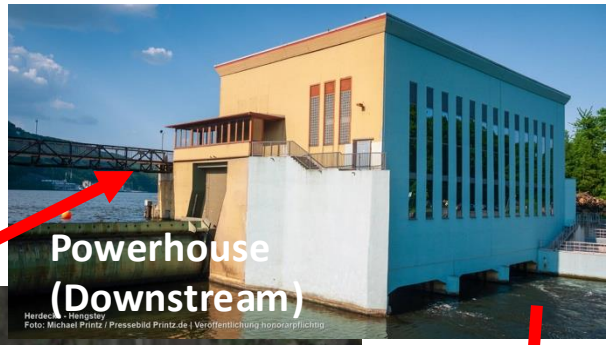


Schemes: Low head in Stream



Hengstey, Power plant, Germany (In Stream)

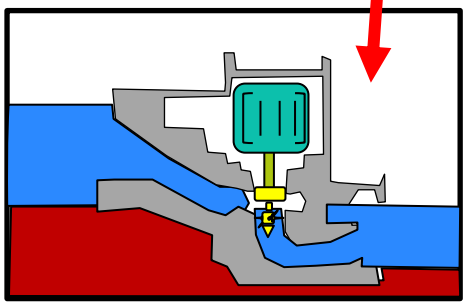






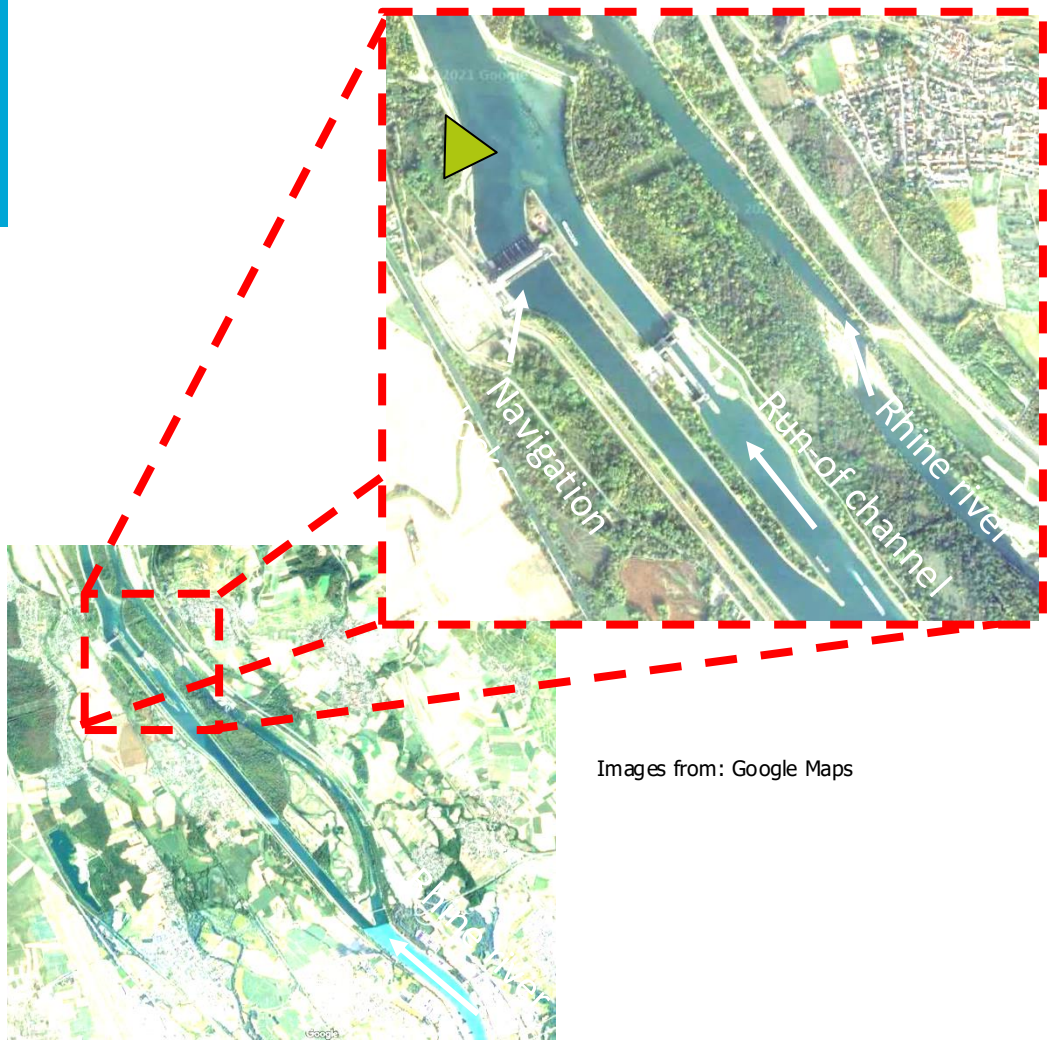


Turbine and generator



Images from:
https://www.komoot.com/highlight/71891

Kembs Power plant, France (Stream Side)



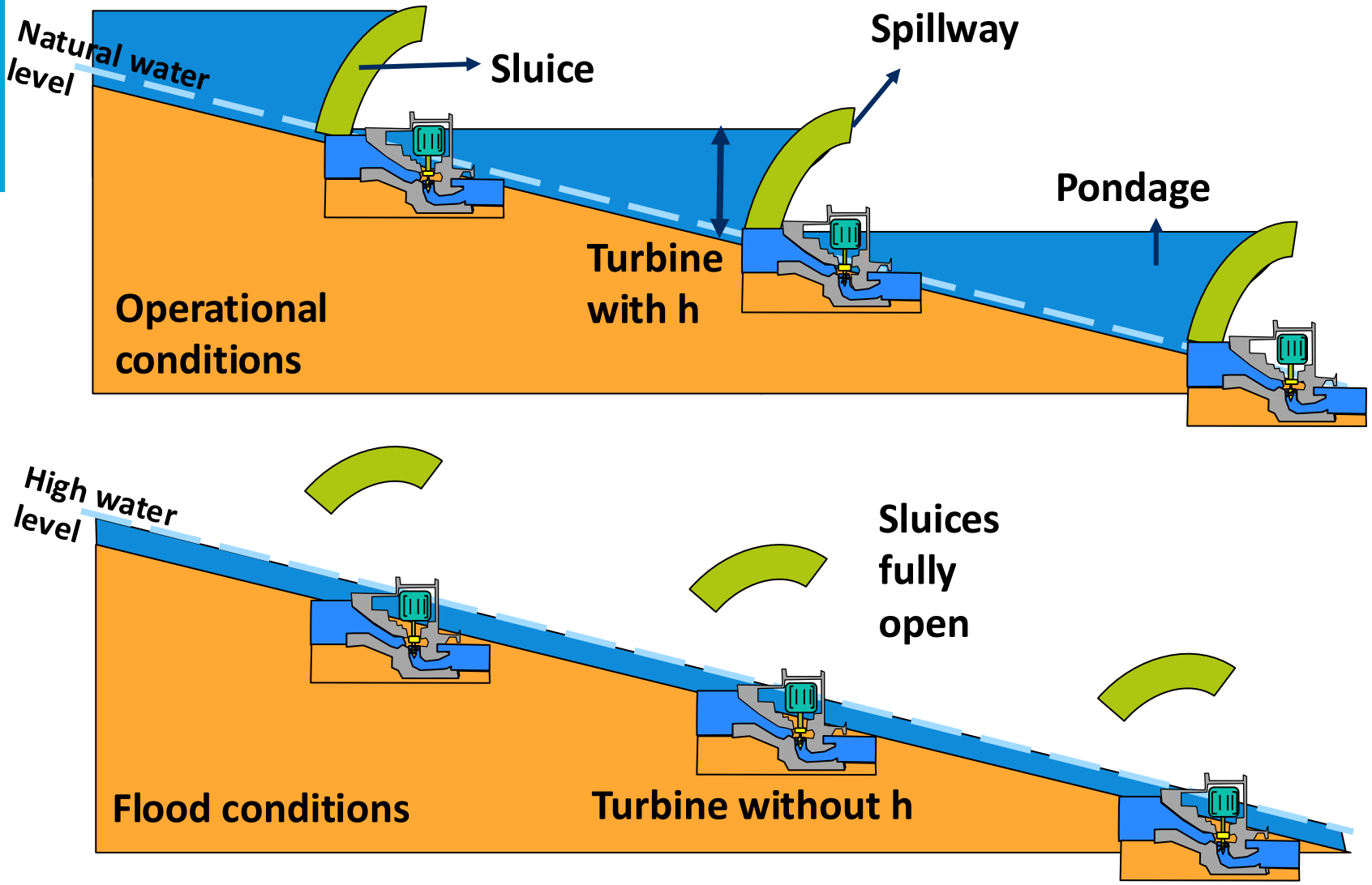
Images from: Google Maps



Downstream view

Image from:
https://openagenda.com/semainedelindustrie2016/events/edf-visite-de-la-centrale-hydroelectrique-de-kembs

Scheme: Run-of-river cascade



Dutch Run-of-river plants (low head)

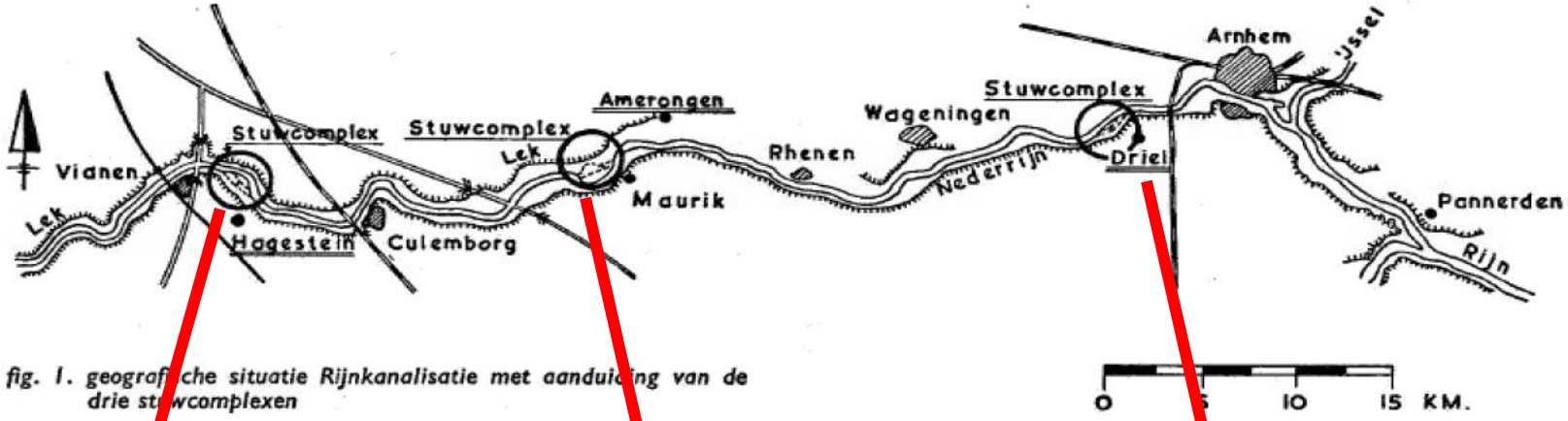
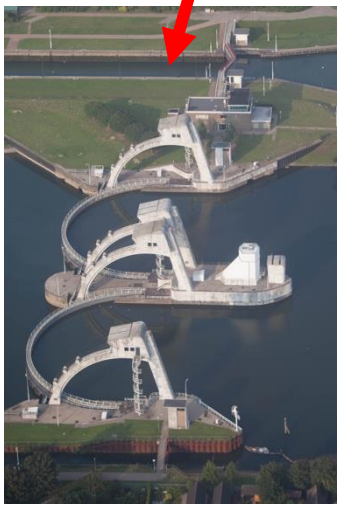
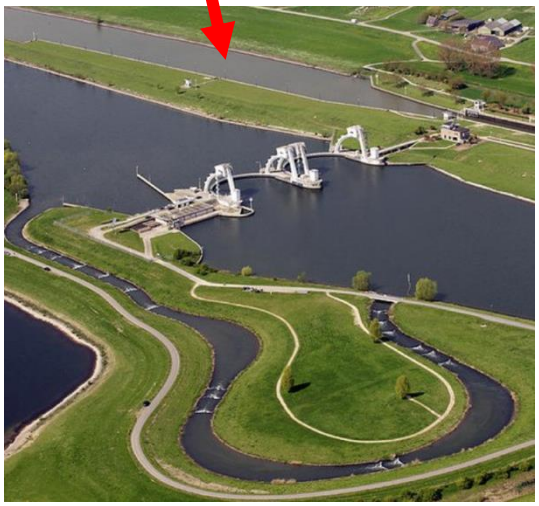


fig. 1. geografische situatie Rijnkanalisatie met aanduiding van de drie stuwcomplexen





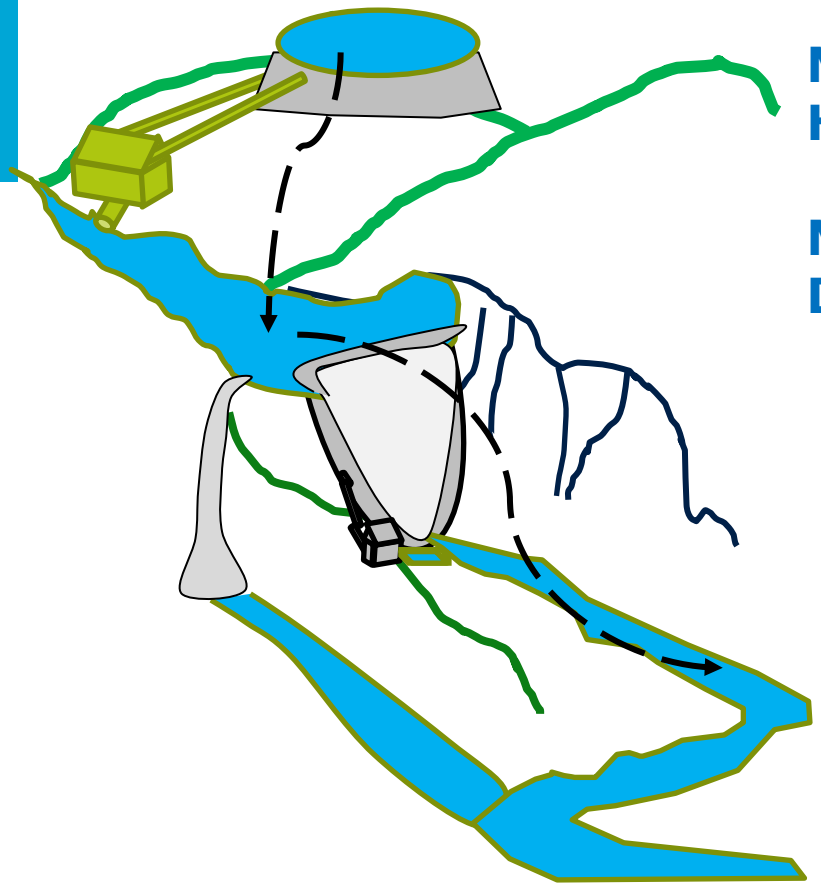


4. Pumped Storage



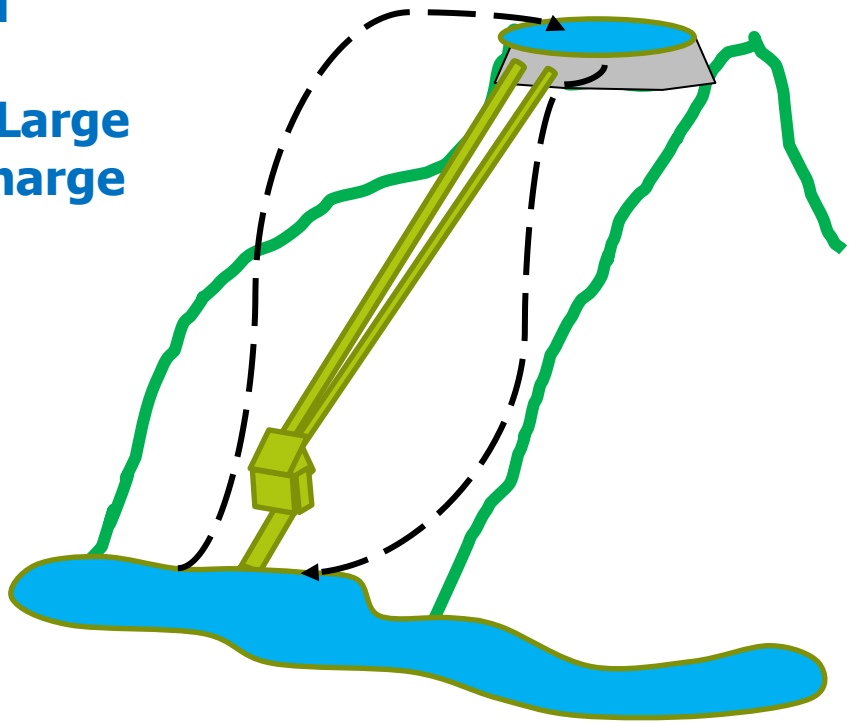


Scheme: Pump back and Off-stream

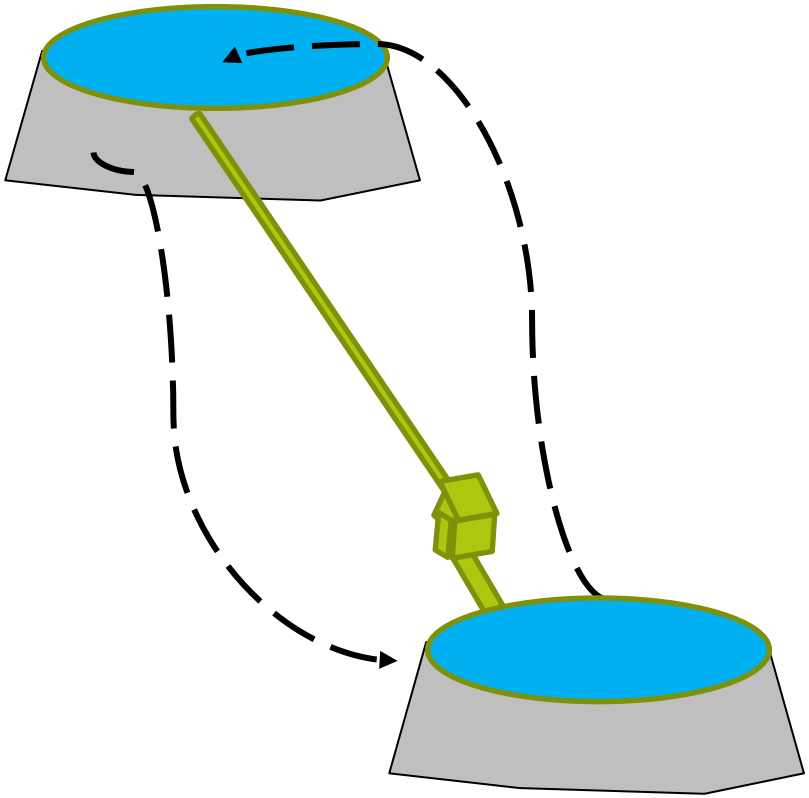


Mid-Large Head

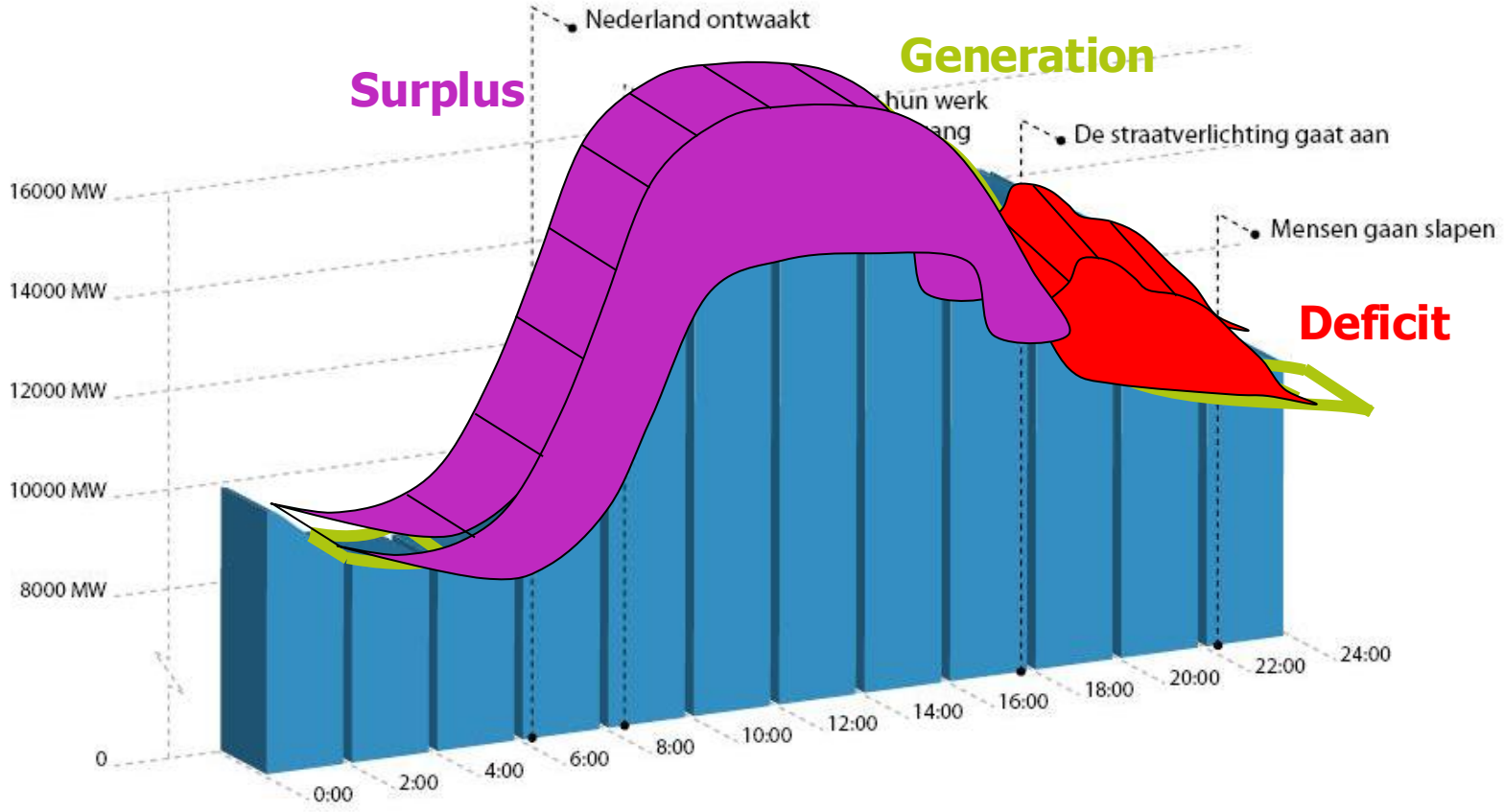
Mid-Large Discharge



Scheme: Closed loop

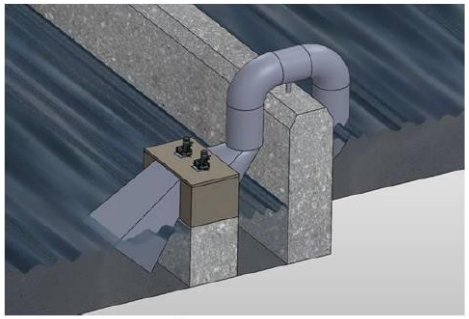


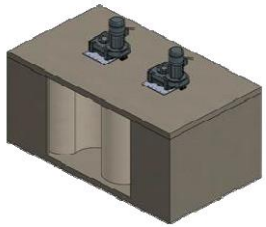
Demand Vs. Capacity



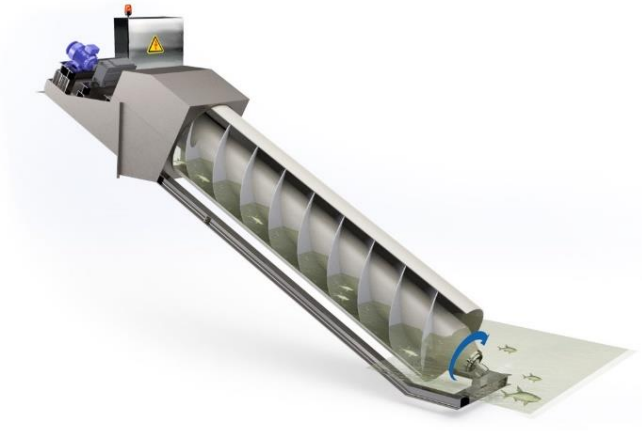
https://sysmod.tbm.tudelft.nl/wiki/index.php/Typologie_van_systeemgedrag

5. Small hydropower (inland)





RONAMIC



Fishflowinnovations.nl



http://www.altenergymag.com/article/2016/04/what-is-hydropower-in-a-pipe/23404/

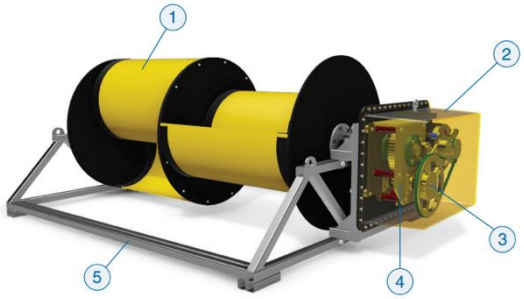


Image Key

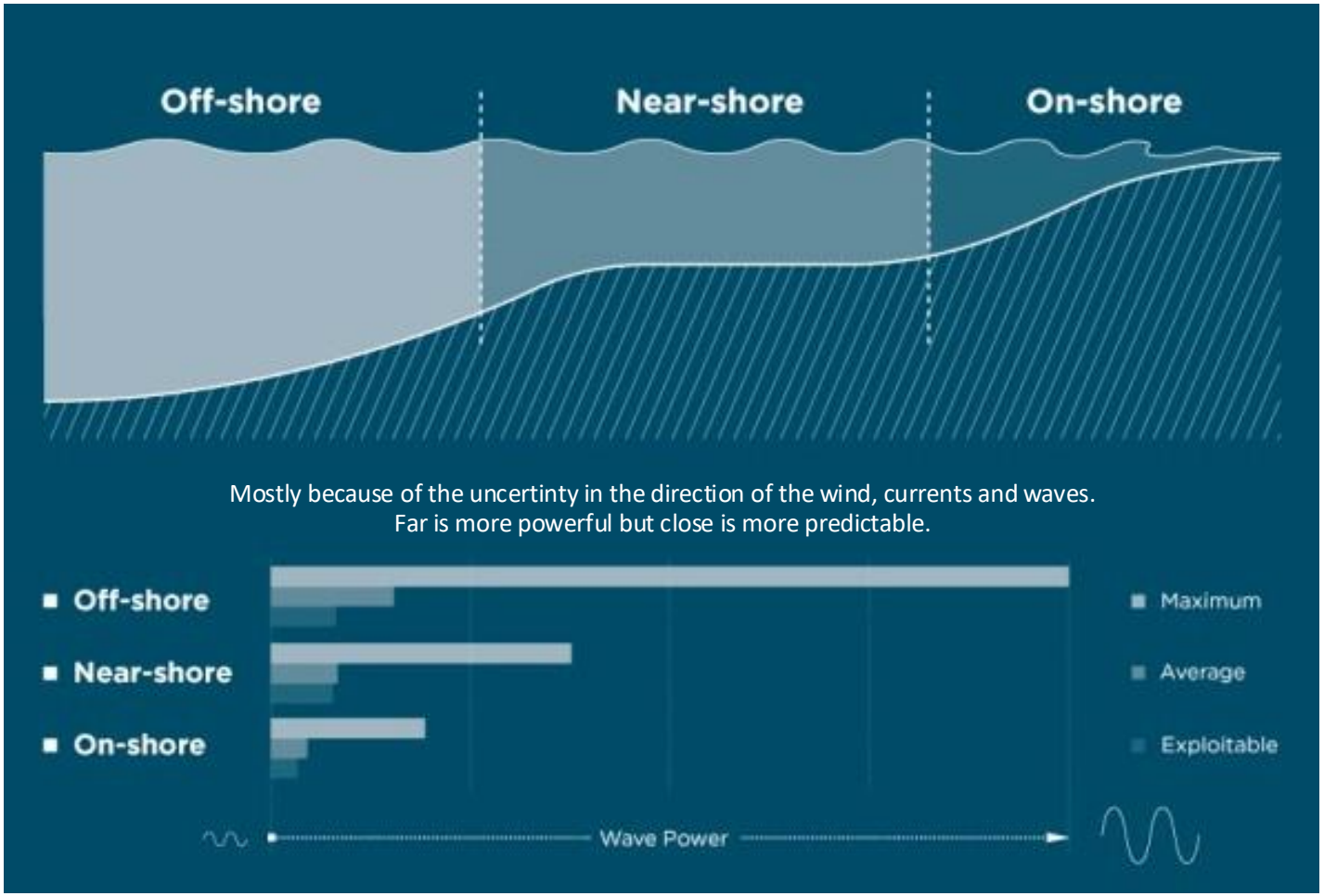
1. Savonius rotor
2. Permanent magnet generator
3. Belt drive
4. Gear box
5. Cradle

Product Specs

- 6' tall
- 13' wide
- 8' deep
- 6,000 lbs dry weight

Hydrovolts.com

Shore wave power relative to location



Source: https://waveenergyconversiontamu15.weebly.com/theory-of-wave-energy–availability.html

6. Tidal Power plants (Onshore)

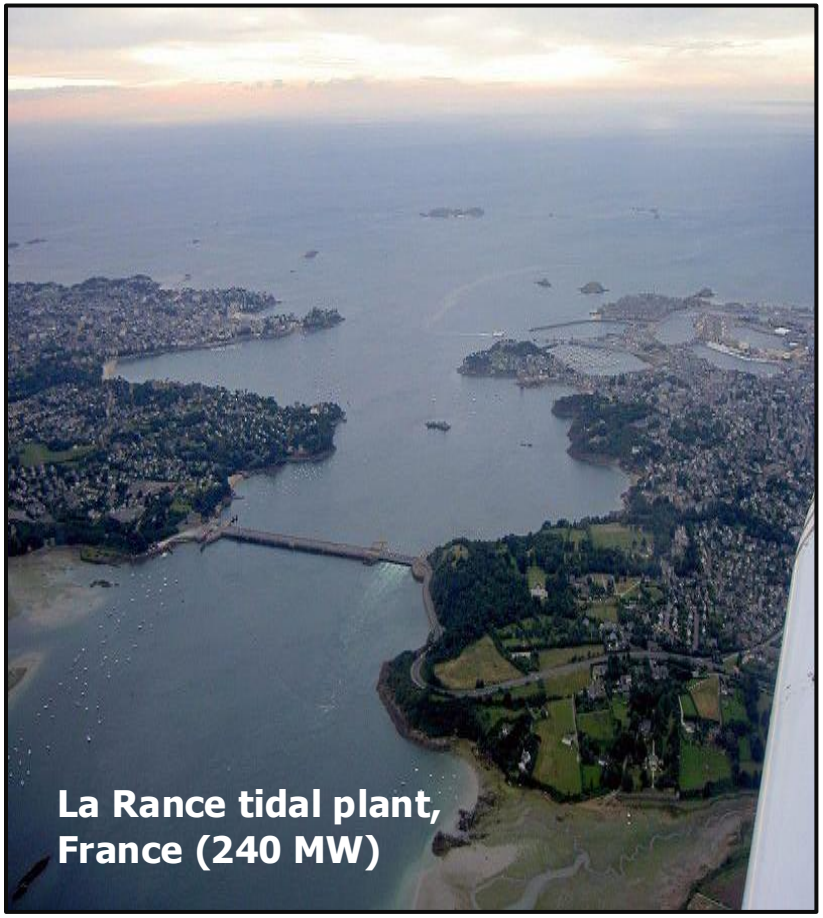
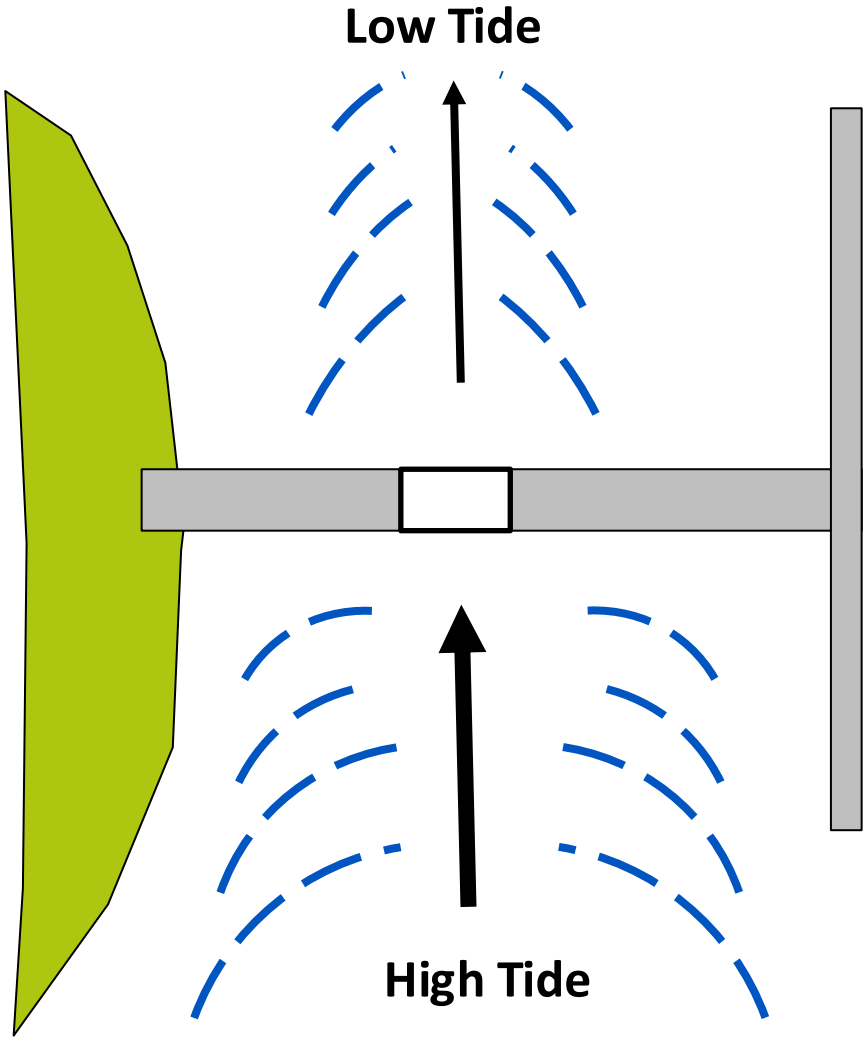


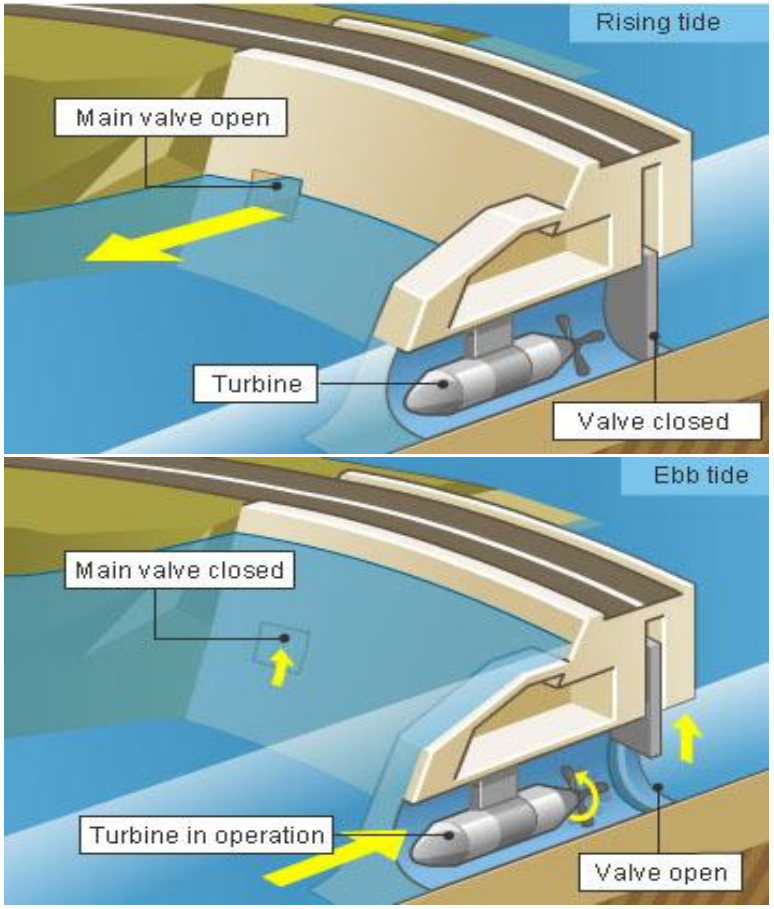
Image from: https://www.erih.net/i-want-to-go-there/site/rance-tidal-power-station



Image from: https://renewablesnow.com/news/nova-manufactures-1st-d2t2-tidal-turbine-to-export-15-units-to-canada-701678/

Schemes: Tidal Barrages (retainment)



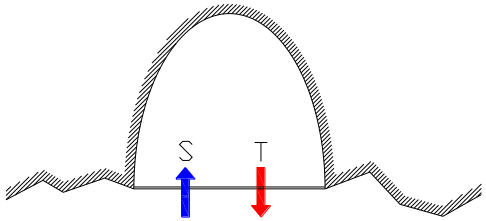


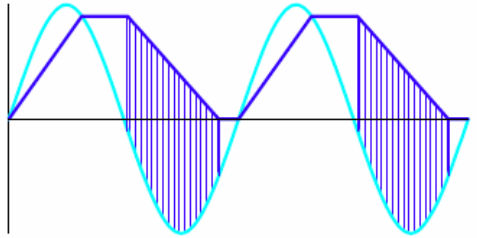
Images from: http://www.worldcolleges.info/Science-Tech/tidal-barrages.php

Tidal Barrage modes

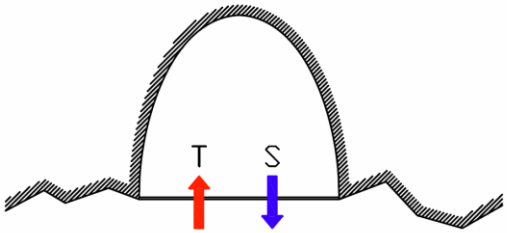
Single effect, seaward

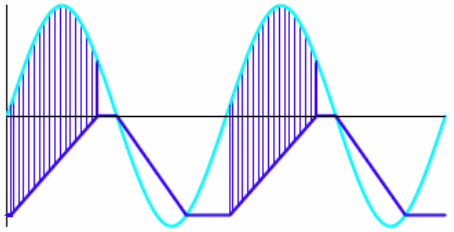
ebb-mode





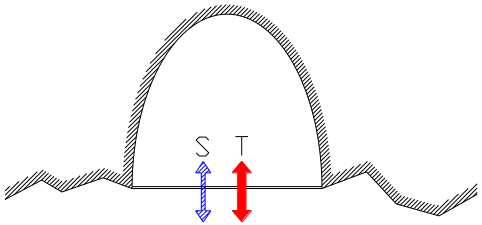
Single effect, flood tide -mode

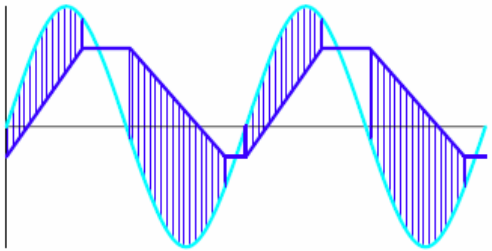




Double effect

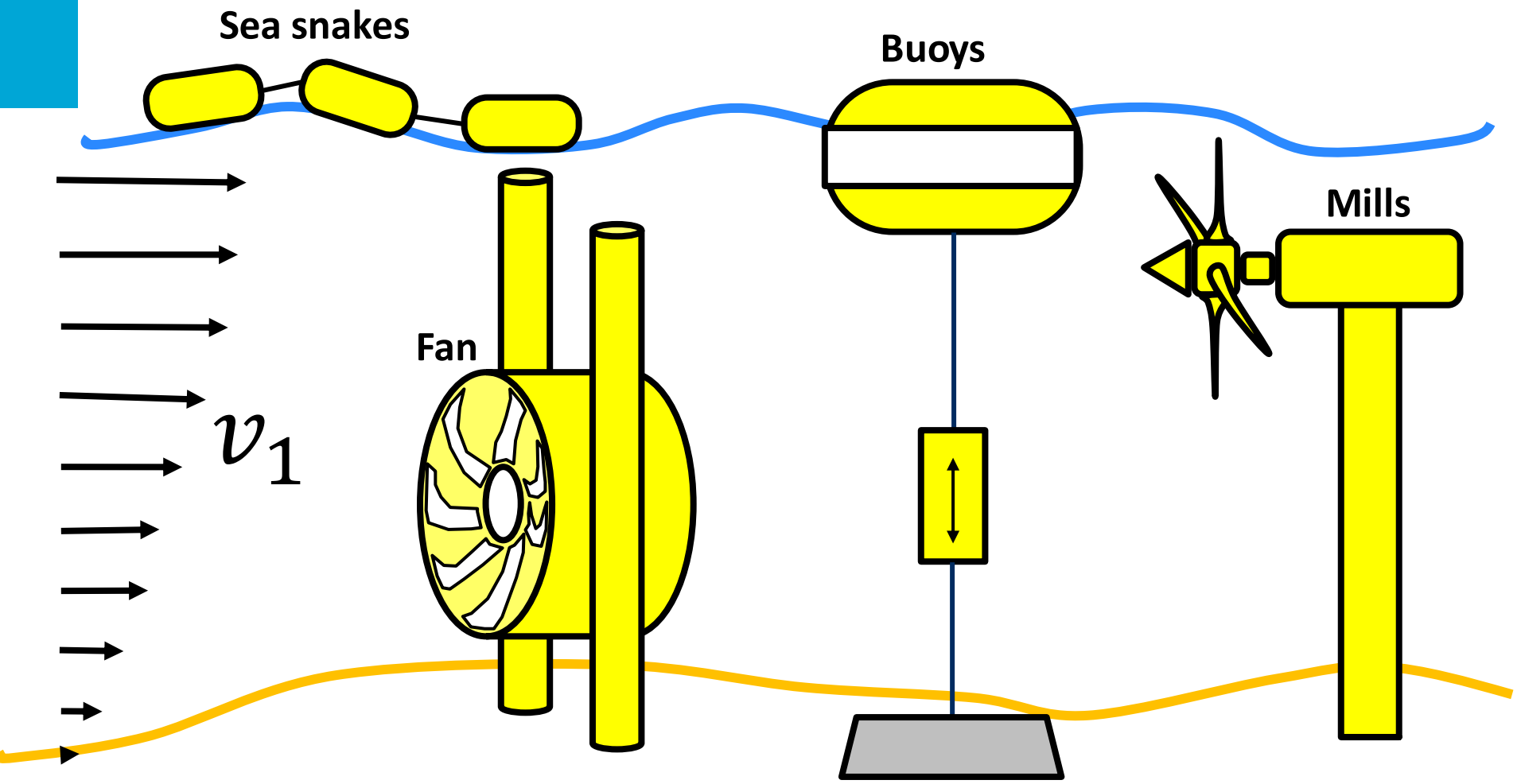
(La Rance)





Schemes: Kinetic turbines and converters

(near and on shore)



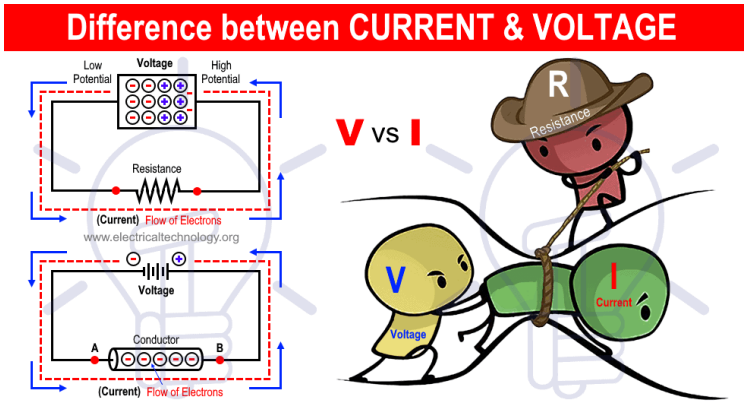
Turbines

Turbines and generators

•**Voltage:** is the potential difference force between two points in an electric field which causes current to flow in the circuit.

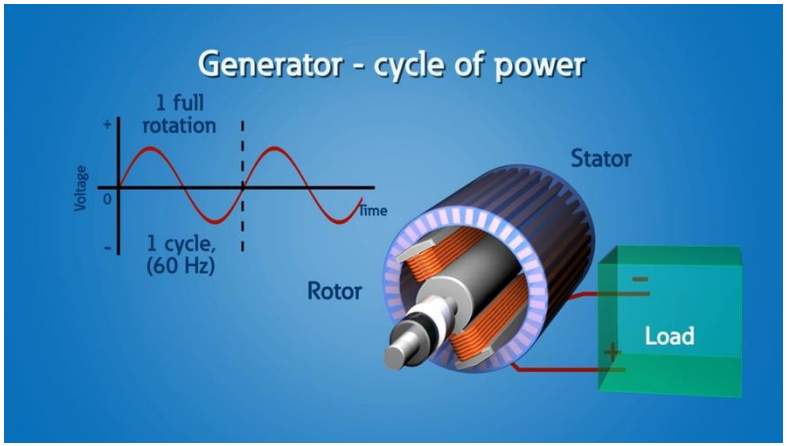
•**Current:** is the rate of charge flow (electrons) between two points caused by voltage.

•**Power:** Is the product between voltage and current ($V \times I$).



- **Mechanical energy** is generated by motion and position of mechanical parts.
- **Electrical energy** is generated by the rate of flow of electrical charges.

$$V \times I = \eta\, Q\, h_{net}\, g\, \rho_w$$



Faraday's Law of Electromagnetic Induction, a change in magnetic flux induces an electric current in the coil.

Types of turbines (kinetic or pressure)

Impulse

(Kinetic energy only at atmospheric pressur

or equal pressure turbines (e.g. Pelton, Turgo, Crossflow…)

Reaction

(Kinetic and pressure energy) or overpressure turbines (e.g. Kaplan, Francis, Bulb…)

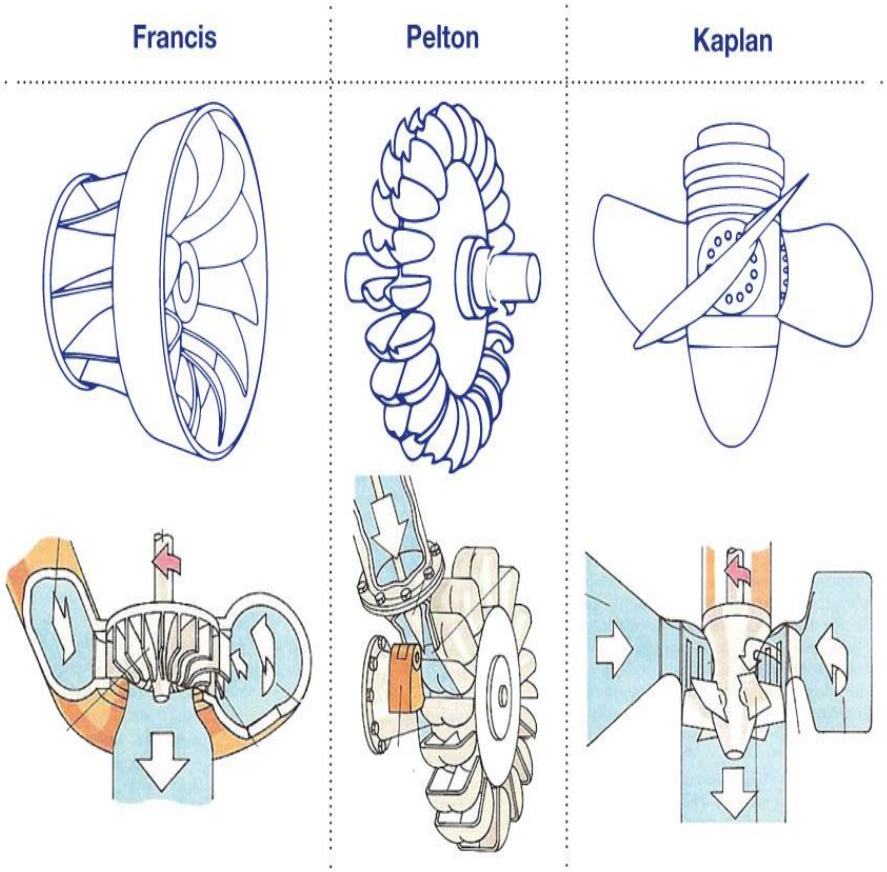


Figure from: http://www.energoconsult.cz/small_hydropower_plants.html

Turbine selection

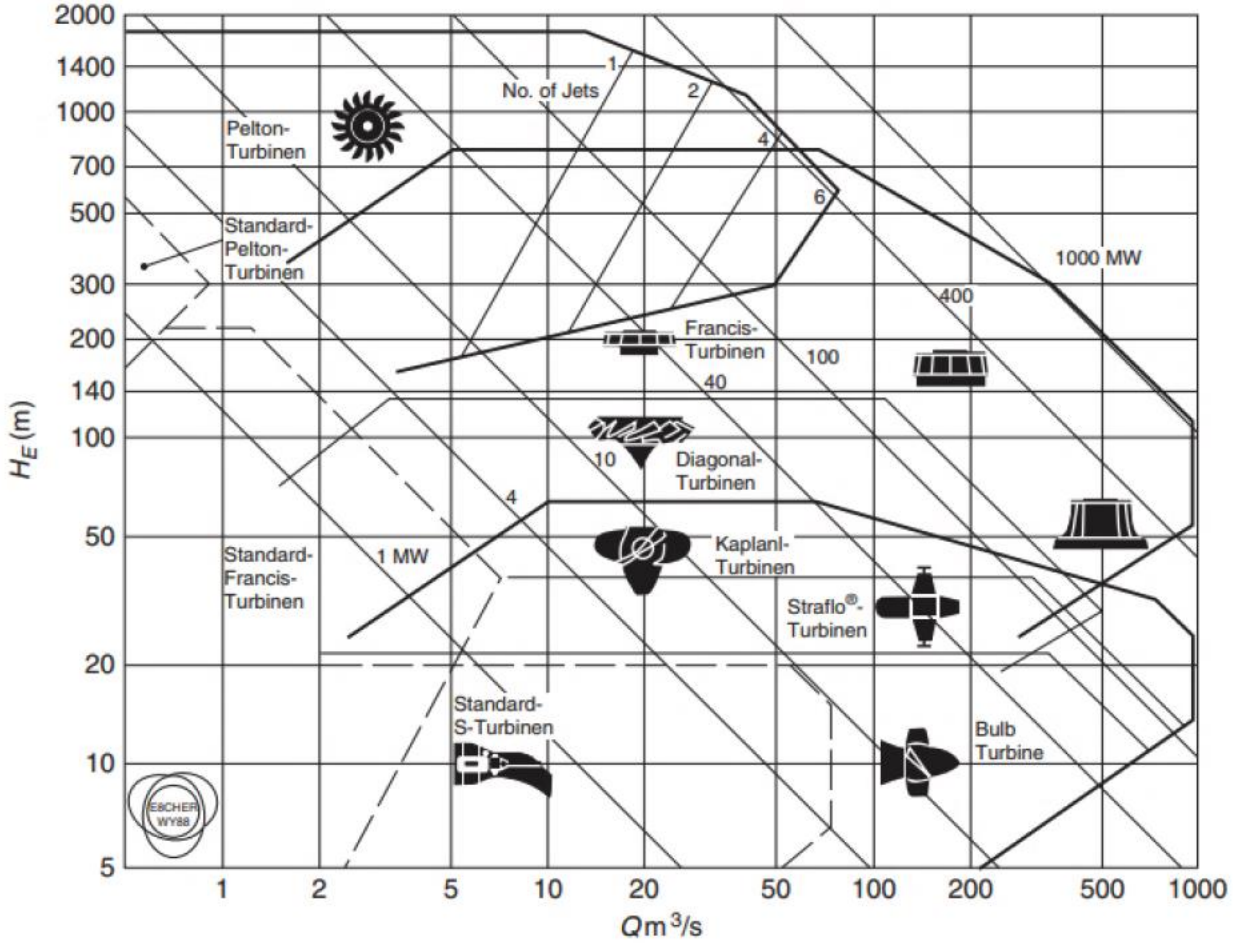


Figure from: Dixon SL and Hall CA (2010), Fluid mechanics and thermodynamics of turbomachinery. Butterworth-Heinemann/Elsevier.

Turbine submergence

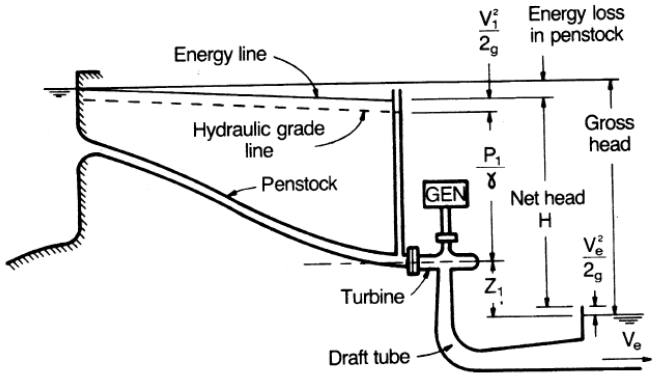
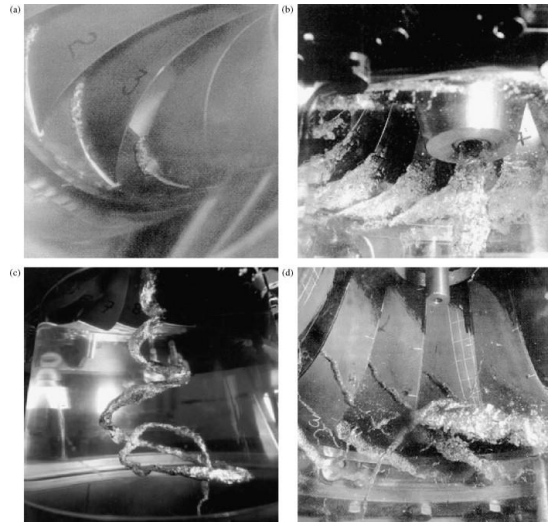


Image form: Ghosh T.K., Prelas M.A. (2011) Hydropower. In: Energy Resources and Systems. Springer, Dordrecht. https://doi.org/10.1007/978-94-007-1402-1_3

|images taken fr0m : Kumar, Pardeep, and R. P. Saini. "Study of cavitation in hydro turbines—A review." *Renewable and Sustainable Energy Reviews* 14.1 (2010): 374-383.



Cavitation cannot always be completely excluded. Hydraulic machines are extremely sensitive to cavitation which usually leads to unacceptable high material deterioration in a very short time.

Regions of high flow velocities and low pressures as around runner blades are conditions under which cavitation may occur.

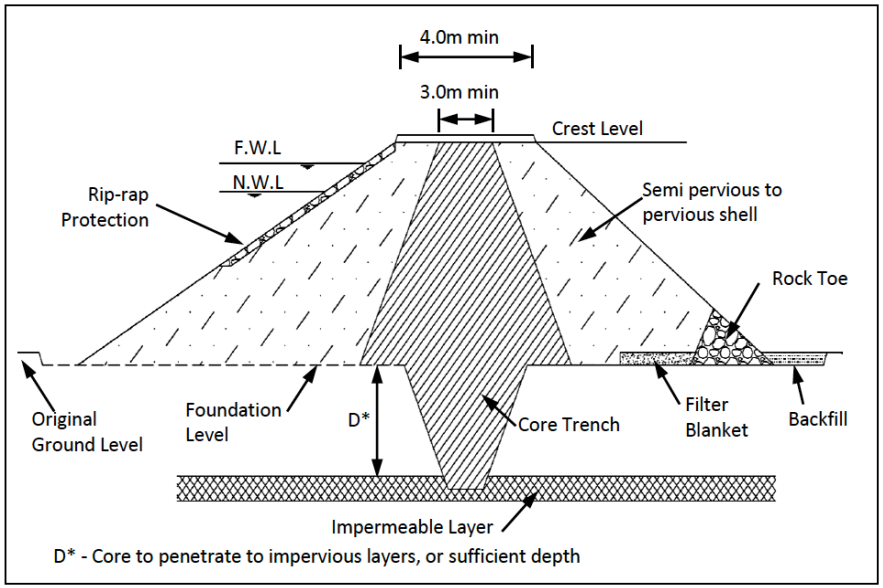
$$h_{S,adm} = h_{at} - h_v - \sigma_c \cdot h_f \qquad h_S \leq (!) h_{S,adm}$$

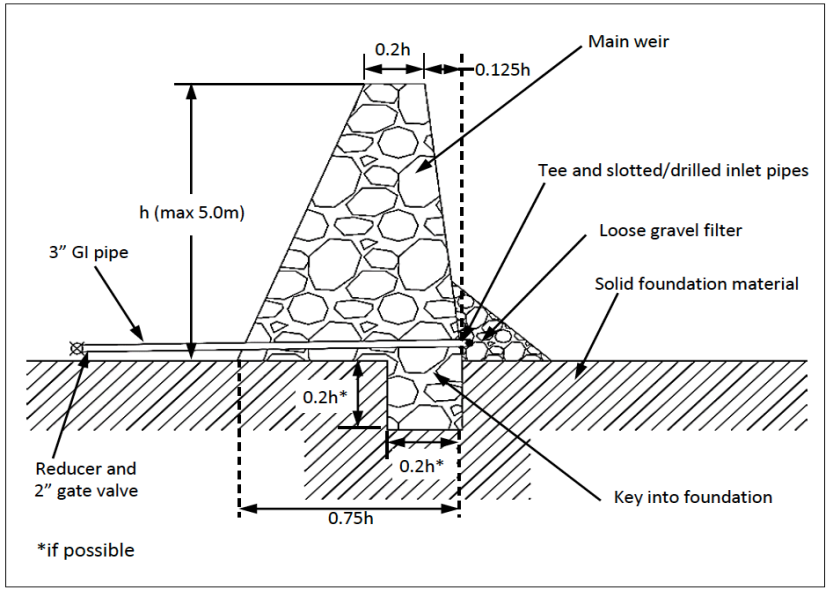
where:

$h_{S,adm}$	admissible draft head [m]
h_f	usable fall head [m]
h_{at}	pressure head of atmospheric pressure [m], at see level h_{at}=10 m
h_v	vapor pressure of water [m]
σ_c	Thoma's cavitation coefficient

Dam design and Safety

Dam Foundations





Figures taken from: http://design-of-small-dams.appspot.com/structures/



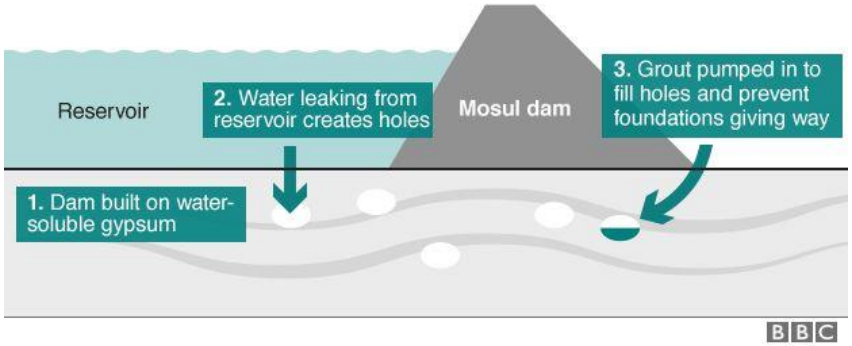


Figure from: https://www.bbc.com/news/world-middle-east-35690616

Dam safety



https://en.wikipedia.org/wiki/Malpasset_Dam



https://en.wikipedia.org/wiki/Teton_Dam



http://web.mst.edu/~rogersda/dams/taum-sauk/watkins-taum-sauk-gsa_branson-compressed.pdf



http://www.latimes.com/local/california/la-me-oroville-spillway-failure-20170216-story.html

Project development and costs

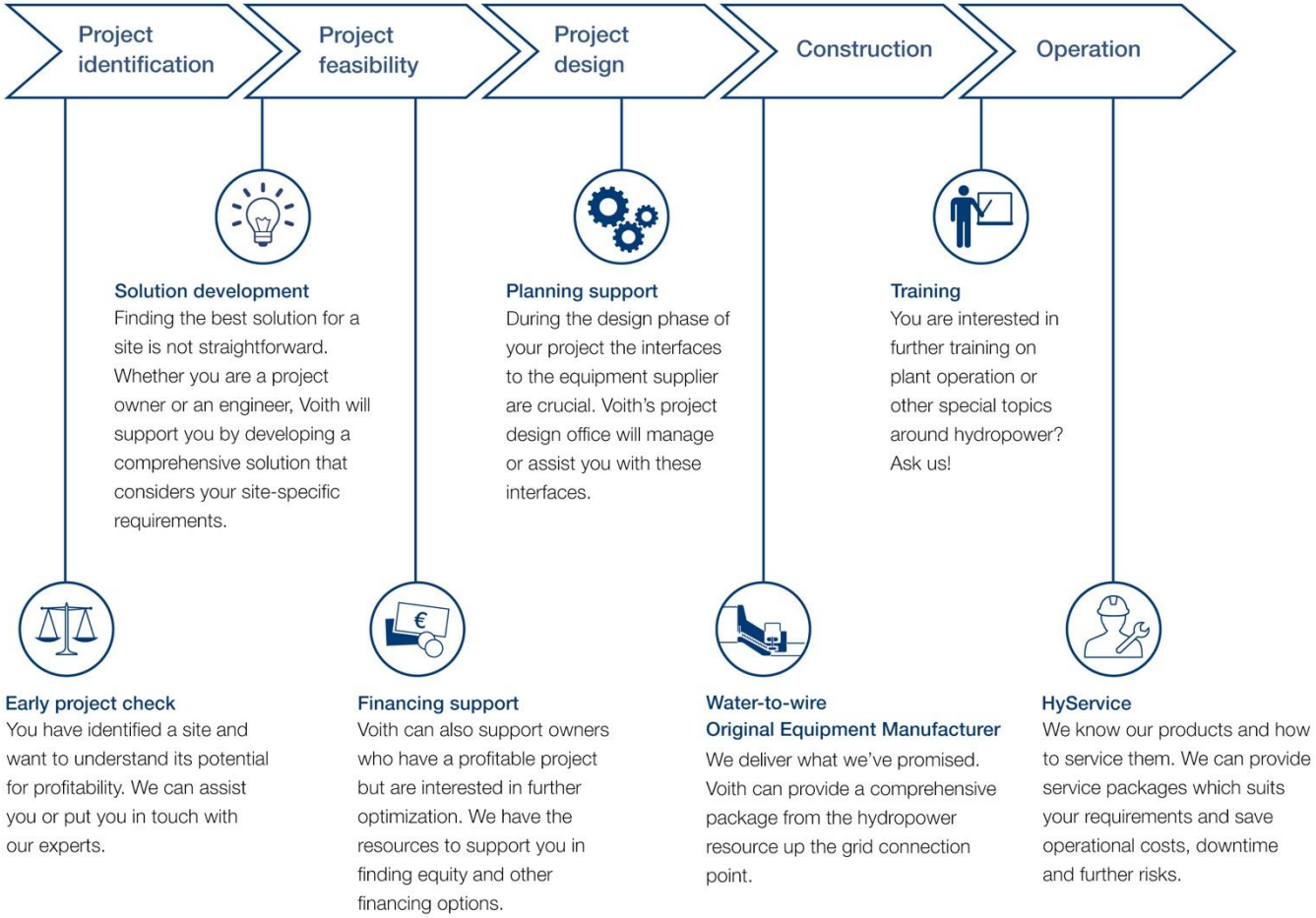


Figure from: https://voith.com/fi-en/hydropower-components/mini-hydro.html

Questions or comments ?



Salvajina dam from upstream view – Suarez, Cauca - Colombia